

SPEAR3 LLRF Upgrade System

System Design Report

RF Department

SSRL / Accelerator Directorate, SLAC National Accelerator Laboratory

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This report is the top-level design reference for the SPEAR3 LLRF Upgrade Project. It describes the system architecture, the scope and high-level design of every subsystem in both the legacy and upgraded configurations, and the interfaces between them. It is the entry point for the detailed engineering design of each subsystem, not a substitute for it: Section 20 indexes the detailed design documents that carry the underlying evidence. The legacy plant is described in full in Designs/tex/docL-legacy-architecture/L_legacy_system_architecture.pdf (Doc L).

Contents

Purpose and Scope	6
1 Executive Summary	6
1.1 Key System Parameters	6
1.2 Upgrade Drivers	6
2 System-Level Architecture	7
2.1 Legacy System Architecture	7
2.2 Upgraded System Architecture	7
2.3 What Stays, What Changes, What Is New	9
3 Physical Layout and Locations	9
3.1 Cabling Between Locations	10
4 RF Plant — Retained Physical Infrastructure	10
4.1 Klystron	10
4.2 Waveguide Distribution	10
4.3 RF Cavities	11
4.4 Drive Amplifier	11
4.5 Legacy Collector Power Protection	11
4.6 RF Signal Monitoring	11
5 Subsystem 1: LLRF Controller	14
5.1 Legacy System	14
5.2 Upgraded System — Dimtel LLRF9/476	14
5.3 Two-Unit Configuration for SPEAR3	15
5.4 Key Performance Specifications	16
5.5 Interfaces	16
6 Subsystem 2: High Voltage Power Supply (HVPS)	16
6.1 Power Section (Retained)	16
6.2 Legacy Controller	17
6.3 Upgraded Controller	18
6.4 HVPS Controller Interface	19
7 Subsystem 3: RF Machine Protection System (RF MPS)	20
7.1 Purpose and Protection Philosophy	20
7.2 Legacy System	21
7.3 Upgraded System — ControlLogix 1756	21
7.4 RF MPS Outputs to Interface Chassis	21
7.5 RF MPS Inputs from Interface Chassis	22
7.6 RF MPS Role in the Protection Chain	22
7.7 EPICS Interface	23
7.8 Interfaces Summary	23
7.9 Source Documents	23

8	Subsystem 4: Interface Chassis	24
8.1	Purpose	24
8.2	Interface Summary	24
8.3	Implementation Status	24
9	Subsystem 5: Personnel Protection System (PPS) Interface	25
9.1	PPS Overview	25
9.2	Legacy PPS Design (Problems)	25
9.3	Upgraded PPS Design	25
9.4	PPS Regulatory Considerations	26
10	Subsystem 6: Tuner Control System	26
10.1	Physical Mechanism	26
10.2	Legacy Controller	27
10.3	Upgraded Controller	27
10.4	Load Angle Offset Loop	27
10.5	Tuner Interfaces	27
10.6	Risk Note	27
11	Subsystem 7: Waveform Buffer System	28
11.1	Purpose	28
11.2	Channel Configuration	28
11.3	Key Features	29
11.4	Interfaces	29
12	Subsystem 8: Arc Detection System	29
12.1	Purpose	29
12.2	Technology	29
12.3	Installation	29
12.4	Signal Path and Design	30
12.5	Permit Logic	31
12.6	Interfaces	31
13	Subsystem 9: Klystron Cathode Heater	31
13.1	Legacy System	31
13.2	Upgraded System	32
13.3	Interfaces	33
14	Subsystem 10: Control Software	34
14.1	Legacy Software	34
14.2	Upgraded Software — EPICS/MATLAB Coordinator	34
14.3	Key Design Principles	34
15	Control Loop Architecture	35
15.1	Fast RF Feedback (Direct Loop)	35
15.2	HVPS Supervisory Loop (~1 Hz)	35
15.3	Tuner Control Loops ($\times 4$, ~1 Hz)	35
15.4	Load Angle Offset Loop (~1 Hz)	36
15.5	Station State Machine	36

15.6 Calibration	37
15.7 Eliminated Loops	37
16 Inter-Subsystem Interface Matrix	37
16.1 Interface Signal Types Summary	39
16.2 Open Interface Questions	39
17 Protection Chain and Interlock Architecture	40
17.1 Protection Philosophy	40
17.2 Fault Propagation Example — RF Arc Event	41
17.3 Fail-Safe Design	41
18 Communication Architecture	42
18.1 EPICS Channel Access Network	42
18.2 Non-EPICS Communication Paths	42
19 Implementation Phases and Risk Summary	43
19.1 Implementation Phases	43
19.2 Status	44
19.3 Key Technical Risks	44
19.4 Success Criteria	45
20 Appendix: Source Document Index	45
20.1 Governing project document	46
20.2 LLRF Source Material	46
20.3 HVPS Source Material	46
20.4 Switchgear and PPS Source Material	47
20.5 Companion design documents	48

List of Figures

1	Upgraded system architecture. The Interface Chassis is the single hub for machine and equipment protection; personnel safety runs on a physically and logically separate path through the PPS Interface Box to the SPEAR PPS.	8
2	RF plant signal flow. Heavy lines are waveguide, light lines coaxial. Circled numbers are the directional-coupler channel indices used in Section 4.6.	14
3	HVPS power chain in Building B514. The 15° phase offset between the two rectifier transformers is what turns two 6-pulse bridges into a single 12-pulse supply.	17
4	Arc detection signal path. The two outputs run in parallel: an all-hardware OR gives the fast trip, while the 10-bit latch preserves which sensor fired for post-mortem analysis. . . .	30
5	Klystron cathode heater controller architecture. Three paths are deliberately separate: the MPS ramps the Chroma output through the <i>analog</i> programming input, so controlled warm-up and cooldown do not depend on the network; the Ethernet link carries setpoints and readback only; and the existing analog V/I meters are retained, so the delivered output is monitored in hardware independently of both. There are no mechanical components anywhere in the control path.	33
6	Fault propagation for an RF arc event. Each step is roughly three orders of magnitude slower than the one above it. Steps 1–3 are pure hardware: protection is complete before any programmable device is involved. Steps 4–6 are diagnostic and supervisory only — no protective action depends on them, so a PLC or IOC failure degrades observability but never safety.	41

Purpose and Scope

This Physical Design Report is the top-level system design reference for the SPEAR3 Low-Level RF (LLRF) Upgrade Project. It describes the overall system architecture, the scope and high-level design of each subsystem in both the current (legacy) and upgraded configurations, and the interfaces and interactions between subsystems.

This document is intended to serve as the entry point for the detailed engineering design of each subsystem. It consolidates physical design information from across the project and provides traceability to the detailed design documents in this project.

1 Executive Summary

The SPEAR3 RF station provides 476.3 MHz RF power to the SPEAR3 storage ring at the Stanford Synchrotron Radiation Lightsource (SSRL). A single klystron, driven at approximately 800 kW, feeds four single-cell RF cavities through a waveguide distribution network. The combined cavity gap voltage is approximately 2.85 MV.

The LLRF Upgrade Project replaces the entire control electronics chain — not merely the low-level RF controller, but also the RF machine protection system (RF MPS), HVPS controller, tuner motor controllers, and supporting infrastructure. The project will also modify several systems in the existing system. An Interface Chassis that is required to centralize and coordinate the interlocks between system elements, will be redesigned and replaced to interact with the elements of the system. The klystron cathode heater power supply will be modernized. An expanded Waveform Buffer System is being built to provide better system diagnostic signals. And a new optical arc detection system will be installed to replace a system that did not perform as expected.

1.1 Key System Parameters

Parameter	Value
RF Frequency	~476.3 MHz
Total Gap Voltage	~2.85 MV (sum of 4 cavities)
Cavity Gap Voltage	~712 kV each
Klystron Power	~1.5 MW rated, ~800 kW typical operating
HVPS Voltage	up to ~90 kV (negative polarity)
HVPS Operating Voltage	~74 kV at 500 mA beam
Drive Power	~29 W nominal
Number of Cavities	4 (single-cell, individual tuners)
LLRF9 Units	2 active, 2 spare (4 purchased)

Source: Gap voltage and beam current data from LLRF9 commissioning measurements (`11rf/tests/11rf9Tests.pdf`, J. Sebek, 2021). Klystron rated power from HVPS hazard documentation (`hvps/documentation/procedures/spear3HvpsHazards.tex`).

1.2 Upgrade Drivers

- **Hardware obsolescence** — Commercial VXI, PLC-5, and SLC-500 modules are at end of life and no longer vendor supported. Custom SLAC LLRF processing modules are also end-of-life and no longer supported by SLAC.

- **PPS compliance** — Legacy design routes PPS wiring through HVPS controller and has a PLC in the safety chain
- **Performance improvement** — Digital FPGA-based feedback (270 ns loop delay) replaces analog processing
- **Diagnostics** — 16k-sample waveform capture, circular buffers, first-fault detection for RF signals, 100 ms of waveform capture for HVPS signals
- **Maintainability** — Modern hardware, comprehensive documentation, EPICS/MATLAB software

2 System-Level Architecture

2.1 Legacy System Architecture

The legacy SPEAR3 LLRF system was originally designed for the PEP-II B-Factory (circa 1997) and later adapted for SPEAR3. It consists of:

- **LLRF Controller:** Custom PEP-II analog RF Processor (RFP) module in a VXI chassis. The VXI crate contains a CPU, an Allen-Bradley controller, a Clock Module, the RFP module (heart of the fast feedback system), three IQA modules (amplitude/phase monitors), and an Arc Interface Module.
- **Control Software:** Six SNL (State Notation Language) programs on VxWorks RTOS, compiled into a single `rfSeq` IOC library:
 - `rf_tuner_loop.st,v` — Cavity tuner stepper motor control
 - `rf_hvps_loop.st,v` — HVPS supervisory control and regulation
 - `rf_states.st,v` — RF station state machine control (PEP-II heritage, R. Sass, 1997)
 - `rf_dac_loop.st,v` — Drive Power and Gap Voltage RFP/GFF DAC loop (S. Allison, 1997)
 - `rf_calib.st,v` — Calibration sequences (R. Claus, SLAC/PEP-II LLRF Group)
 - `rf_msgs.st,v` — Message logging and CAMAC TAXI error monitoring
 - See: `spear-rf-code-legacy/rfApp/src/seq/Makefile,v`, `spear-rf-code-legacy/rfApp/src/seq/rf_dac_loop_pvs.h,v`
- **HVPS Controller:** Allen-Bradley SLC-500 PLC (processor: AB-1747-L532, scanner: AB-1747-DCM) with Enerpro SCR gate driver boards, housed in a Hoffman NEMA enclosure in Building B118. It communicates with the VXI crate via an Allen-Bradley serial interface to receive commands and to send readbacks and statuses. It communicates with an interface chassis via fiber optic signals to receive hardware permits and send a hardware status.
 - See: `pps/HoffmanBoxPPSWiring.docx`
- **RF Machine Protection System (RF MPS):** Allen-Bradley PLC-5 (1771 series), (since converted to ControlLogix 1756 with the hardware assembled, software written, tested without RF power). Scope: klystron/RF station equipment protection only — distinct from the facility-wide SPEAR MPS which provides an external permit input.
 - See: `llrf/documentation/mpsWiringDiagrams/`, `hvps/architecture/designNotes/RFSystemMPSRequirements.docx`
- **Tuner Motor Controllers:** Allen-Bradley 1746-HSTP1 stepper modules with Superior Electric SS2000MD4-M Slo-Syn PWM drivers (obsolete); stepper motors are Superior Electric Slo-Syn M093-FC11 (NEMA 34D)
 - See: `llrf/tuners/SLO-SYN_SS2000MD4M_Step_Drive_Translator_Manual.pdf`, `llrf/tuners/SLO-SYN.pdf`
- **Interlock System:** Distributed across analog modules, PLC I/O, and direct wiring. Some of the wiring is connected in a Local Control Chassis, which connects to the Fast Interlock Chassis. The interlock chassis also has fiber optic inputs to detect potential arcs in the waveguide system. The interlock chassis summarizes the status of these interlocks and reports them back to the VXI system through an Arc Interlock Module, located in the VXI crate.
- **Communication:** The main intelligence for the existing system is a Kinetics Systems VXI IOC in the system VXI crate. The crate has a VME Allen-Bradley communication module through which it communicates, via a serial link, with the two Allen-Bradley RF MPS crates, the four cavity tuner stepper motor controllers, and the HVPS controller.
 - See: `spear-rf-code-legacy/rfApp/src/seq/rf_msgs.st,v`

2.2 Upgraded System Architecture

The upgraded system replaces all control electronics while retaining the RF plant physical infrastructure (klystron, cavities, waveguide, HVPS power section, tuner mechanical assemblies). The upgraded architecture is:

Key Architectural Principle: The Interface Chassis implements **machine/equipment protection and operational interlocks** (LLRF/HVPS/RF MPS coordination), while personnel safety (PPS) functions are implemented exclusively in a completely separate, dedicated **PPS Interface Box**. These two safety-related subsystems are architecturally independent:

- **Operator Layer:** has relative little control with only necessary readbacks.
- **Expert Layer:** with details used by system experts to control and read data from the system. There will be MATLAB tools, written by Dmitry for the LLRF9 and written in-house for other systems.

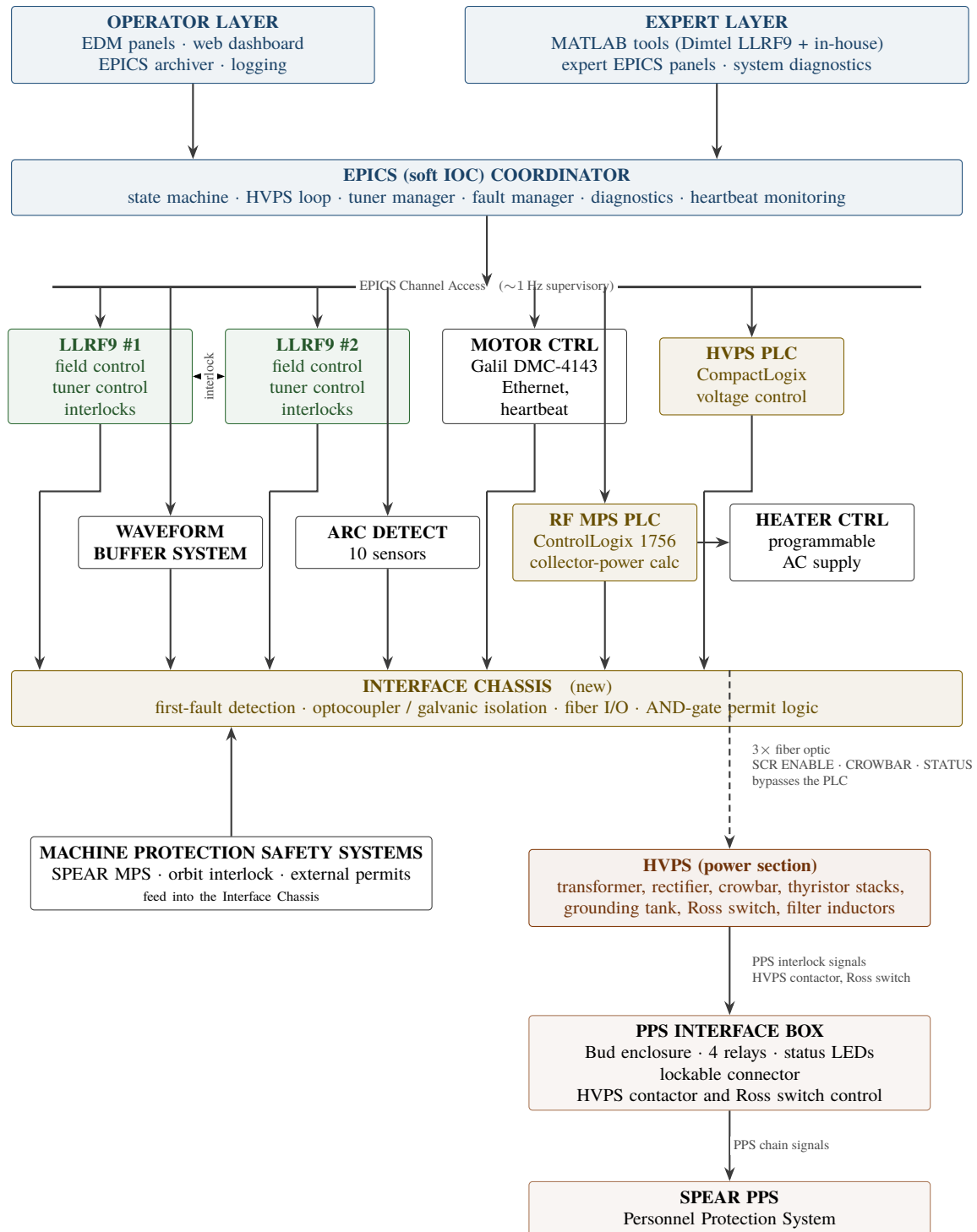


Figure 1: Upgraded system architecture. The Interface Chassis is the single hub for machine and equipment protection; personnel safety runs on a physically and logically separate path through the PPS Interface Box to the SPEAR PPS.

- **HVPS Power section** has three fiber optic cables between Interface Chassis.
- **Heater Control:** will be controlled by RF MPS system.
- **Interface Chassis (Machine/Equipment Protection):** First-fault detection, optocoupler and/or other galvanic isolation, fiber I/O for LLRF9/HVPS/RF MPS/orbit coordination and other machine-protection interlocks
- **PPS Interface Box (Personnel Safety / PPS):** Separate Bud enclosure with 4 relays, status LEDs, PPS-lockable connector for HVPS contactor relay and Ross switch control

2.3 What Stays, What Changes, What Is New

Retained Physical Hardware (same equipment):

- Klystron and its RF output
- 4 RF cavities and waveguide distribution (circulator, magic-tees, waveguide loads)
- Stepper motors (M093-FC11 or equivalent) and mechanical tuner assemblies
- Linear potentiometers on tuners (position indication)
- HVPS power electronics (transformer, rectifier, oil system, crowbar)
- Vacuum contactor (Ross HQ3) and contactor controller (Ross HCA-1-A)
- Ross grounding switch, Danfysik DC-CT, Pearson CT-110
- Field cabling (Belden 83715 15C #16 Teflon, Belden 83709 9C #16 Teflon)

Replaced / Upgraded:

- LLRF Controller: Analog RFP → Dimtel LLRF9/476 (×2 units)
 - RF MPS: PLC-5 1771 → ControlLogix 1756
 - HVPS Controller: SLC-500 → CompactLogix PLC + new Enerpro board + redesigned analog regulator
 - Tuner Motor Controllers: AB 1746-HSTP1 + Slo-Syn (**in service**) → Galil DMC-4143 Rev 1.3h 4-axis motion controller (**in development — not installed, not operating; goes online when the LLRF upgrade completes**) (commissioned August 2025, operational)
 - See: `llrf/tuners/galil/functioningGalil20250825SwapABToManual.txt`, `llrf/tuners/galil/firstMotion2024.txt`
 - Control Software: SNL/VxWorks → EPICS + LLRF9 internal EPICS IOC + MATLAB applications
 - Operator Interface: Legacy EDM panels → modernized panels
- New Subsystems** (did not exist in legacy):
- Interface Chassis — central hardware interlock coordination hub
 - Waveform Buffer System — 8 RF + 4 HVPS channel extended monitoring
 - Arc Detection — 10 Microstep-MIS optical sensors across 6 receiver units (5 active, 1 spare): 4 cavity windows (2 sensors per window on 4 receivers), 1 klystron window + 1 circulator (sharing 1 receiver), 1 spare receiver
 - PPS Interface Box
- Enhanced Subsystems** (upgraded from legacy):
- Klystron Heater Controller — Motor-driven variac → Chroma programmable AC supply; MPS controls via analog ramp, EPICS monitors/enables via Ethernet (ASYN VISA ASCII module).
 - Collector Power Protection — Forward power proxy → Waveform Buffer System measures HVPS voltage and current and klystron forward power → Collector power calculation performed in RF MPS

3 Physical Layout and Locations

The SPEAR3 RF station spans multiple buildings and locations at SSRL:

Location	Equipment	Notes
Building B118 (Power Supply Room)	HVPS Controller (Hoffman Box)	HVPS control location
Building B514 (HVPS Substation)	HVPS Main Tank (transformer, rectifier, inductor, filter caps), Phase Tank (12 thyristor stacks), Crowbar Tank (4 thyristor stacks, output V-divider)	High-voltage power section; FR3 oil-filled, N ₂ blanket
Contactor Disconnect Panel (Switchgear, adjacent to B514)	Vacuum contactor (Ross HQ3), Contactor controller (Ross HCA-1-A), K4/MX/RR/L1 relays, S5 auxiliary contact	12.47 kV AC switchgear

Location	Equipment	Notes
Termination Tank (B132, near klystron)	HV cable termination, Ross Engineering HV relay, Danfysik DC-CT, Pearson CT-110	Mineral oil filled
Switch-over Tank (adjacent B514)	HV cable connections between HVPS1/HVPS2 and klystron	FR3 oil filled
Building B132 (Klystron)	Klystron, drive amplifier, LLRF9 units, RF MPS PLC, Interface Chassis, Waveform Buffer, Motion Controller, Arc Detector Chassis, soft IOC (may be in B137 — TBD confirmed with controls)	Main control electronics location
SPEAR3 Storage Ring Tunnel	4 RF cavities, waveguide distribution, tuner assemblies, arc detection sensors	Radiation area

3.1 Cabling Between Locations

Cable Run	Cable Type	Conductors	Route
B118 → Switchgear (Contactor)	Belden 83715	15C #16 Teflon	TS-5 to contactor controller
B118 → Termination Tank (Grounding)	Belden 83709 + Belden 83715	9C + 15C #16 Teflon	TS-6 to grounding tank
B118 → B514 (HVPS)	Electrical cable pairs	SCR trigger cables (12 pairs)	Controller to Phase Tank thyristor stacks
B118 → B514 (HVPS)	Fiber optic	SCR ENABLE, CROWBAR, STATUS	Controller to HVPS (upgrade: via Interface Chassis)
B132 → Tunnel	Coax cables	RF signals (forward, reflected, probe). Similar forward, reflected power from klystron and circulator load with B132	LLRF inputs from cavities
B132 → Tunnel	Multi-conductor	Motor power + encoder signals	To tuner assemblies

4 RF Plant — Retained Physical Infrastructure

The RF plant is the physical chain that delivers 476.3 MHz RF power from the klystron to the storage ring beam. All of this hardware is retained as-is during the upgrade.

4.1 Klystron

The single klystron is located in Building B132 and operates at approximately 800 kW output power, driven at ~29 W input from the drive amplifier. The klystron cathode is powered by the HVPS at up to ~90 kV (nominal ~74.7 kV at 500 mA beam current). It has a non-full-power collector requiring dedicated protection (see Section 11.3 for upgrade and current implementation in Section 4.5).

4.2 Waveguide Distribution

The klystron output feeds a waveguide network consisting of:

- A circulator (protects klystron from reflected power)
- Magic-tee power splitters (two stage splitter that distributes power to 4 cavities)
- Waveguide loads (absorb rejected power)

4.3 RF Cavities

Four single-cell cavities at 476 MHz, each contributing ~712 kV gap voltage for a total of ~2.85 MV. Each cavity has:

- A cavity probe (monitors internal field amplitude and phase)
- A forward power coupler
- A reflected power coupler
- An individual stepper motor tuner (adjusts resonant frequency via mechanical plunger)
- Waveguide window viewports on either side of a ceramic window (for arc detection sensor mounting)

4.4 Drive Amplifier

The drive amplifier is located in Building B132 near the klystron and boosts the LLRF9 output signal (~0 dBm) to ~29 W to drive the klystron input. In the upgrade, the LLRF9 Unit 1 output (from the thermally stabilized output chain on Board 1 or Board 2) connects to the drive amplifier input via coax cable. A KAW2051M12 amplifier datasheet is available in `11rf/driveAmp/`.

4.5 Legacy Collector Power Protection

The current SPEAR3 system includes collector power protection implemented through the HVPS control loop. The legacy system monitors klystron forward power and compares it against configurable limits:

Current Implementation:

- **Monitoring:** `LLRF:HVPS:PCOLL_MAX` PV sets maximum collector power limit
- **Protection Logic:** HVPS loop (`rf_hvps_loop.st,v`) monitors `klystron_forward_power` vs. `max_klystron_forward_power`
- **Action:** When klystron forward power exceeds limits, HVPS voltage is reduced (`delta_proc_voltage_down`)
- **Integration:** Collector limit displayed in HVPS EDL panel (“COLLECTOR LIMIT”)

Note: In addition to the software protection above, there is an MPS limit (responding on the order of ms) that removes the permit from the HVPS if the limit is exceeded. The fast interlock chassis has an internal RF detector module that takes a coupled signal from the klystron forward power. The AB controller monitors HVPS V and I and performs the arithmetic, removing the MPS permit if collector power exceeds the allowable value. Removal of the permit turns off both the HVPS and the drive power.

Legacy Limitations:

- No direct DC power measurement

4.6 RF Signal Monitoring

The amplitudes of the RF data need to be acquired. The RF signals that are currently monitored are listed in the table below, along with their location in the RF plant signal flow.

4.6.1 Monitored RF Signals

#	Signal	Monitored By	Notes
1	Klystron Output Forward Power	LLRF9 Unit 2 BRD2	RF power at the klystron output traveling toward the circulator; primary measure of station RF output level.
2	Klystron Output Reflected Power	LLRF9 Unit 2 BRD2	RF power reflected back into the klystron output port from the load chain; should be near zero; used for klystron protection.
3	Circulator Load Forward Power	LLRF9 Unit 1 BRD3	Power delivered to the circulator load (circulator port 3); absorbs power the circulator does not properly direct to the cavities plus any net reflection returning from the waveguide network. Sits at ~20 kW, increases to ~40 kW at full power with no beam — more significant signal than WG Load 1 Fwd.
4	Circulator Load Reflected Power	Waveform Buffer Ch2	Power reflected from the circulator load back into the circulator; indicates load match quality.

#	Signal	Monitored By	Notes
5	Klystron Drive Power	LLRF9 Unit 2 BRD1	Input drive signal level to the klystron from the drive amplifier; sets the klystron gain and operating point.
6	Station Reference Power	Waveform Buffer Ch3	Monitor of the 476 MHz reference signal level distributed to the LLRF system; loss of this signal indicates a reference chain fault.
7	Waveguide Load 1 Forward Power	Waveform Buffer (slow)	Power entering WG Load 1 at port 4 of Magic Tee 1. Rarely exceeds 2 kW — suitable for slow monitoring only.
8	Waveguide Load 1 Reflected Power	Waveform Buffer Ch4	Power reflected from WG Load 1 back into Magic Tee 1; indicates load match quality.
9	Cavity A Forward Power	LLRF9 Unit 1 BRD1	RF power traveling into Cavity A from Magic Tee 2; used to monitor power distribution balance and provide reference phase for cavity probe phase measurement.
10	Cavity A Reflected Power	LLRF9 Unit 2 BRD3	Power reflected from Cavity A back into the waveguide; elevated reflection indicates cavity detuning or mismatch.
11	Cavity B Forward Power	LLRF9 Unit 1 BRD2	RF power traveling into Cavity B from Magic Tee 2; compared with Cavity A forward to verify magic-tee balance and provide reference phase for cavity probe phase measurement.
12	Cavity B Reflected Power	LLRF9 Unit 2 BRD3	Power reflected from Cavity B; monitored for detuning and arc detection.
13	Cavity A Probe Signal	LLRF9 Unit 1 BRD1	Electric field sampled by Cavity A's internal pickup probe; direct measure of the accelerating voltage amplitude and phase; primary LLRF fast feedback and tuner control input for Cavity A.
14	Cavity B Probe Signal	LLRF9 Unit 1 BRD1	Electric field from Cavity B's internal probe; used in the LLRF vector-sum feedback and Cavity B tuner control.
15	Waveguide Load 2 Forward Power	LLRF9 Unit 1 BRD3	Power entering WG Load 2 at port 4 of Magic Tee 2; absorbs the sum of equal reflected power from Cavities A and B.
16	Waveguide Load 2 Reflected Power	Waveform Buffer Ch5	Power reflected from WG Load 2; indicates match quality of the load.
17	Cavity C Forward Power	LLRF9 Unit 1 BRD2	RF power traveling into Cavity C from Magic Tee 3.
18	Cavity C Reflected Power	LLRF9 Unit 2 BRD3	Power reflected from Cavity C; monitored for detuning and arc detection.
19	Cavity D Forward Power	LLRF9 Unit 2 BRD1	RF power traveling into Cavity D from Magic Tee 3.
20	Cavity D Reflected Power	LLRF9 Unit 2 BRD2	Power reflected from Cavity D.
21	Cavity C Probe Signal	LLRF9 Unit 1 BRD2	Electric field from Cavity C's internal probe; used in the LLRF vector-sum feedback and Cavity C tuner control.

#	Signal	Monitored By	Notes
22	Cavity D Probe Signal	LLRF9 Unit 2 BRD1	Electric field from Cavity D's internal probe; used in the LLRF vector-sum feedback and Cavity D tuner control.
23	Waveguide Load 3 Forward Power	LLRF9 Unit 1 BRD3	Power entering WG Load 3 at port 4 of Magic Tee 3; absorbs the sum of equal reflected power from Cavities C and D.
24	Waveguide Load 3 Re- flected Power	Waveform Buffer Ch6	Power reflected from WG Load 3; indicates match quality of the load.

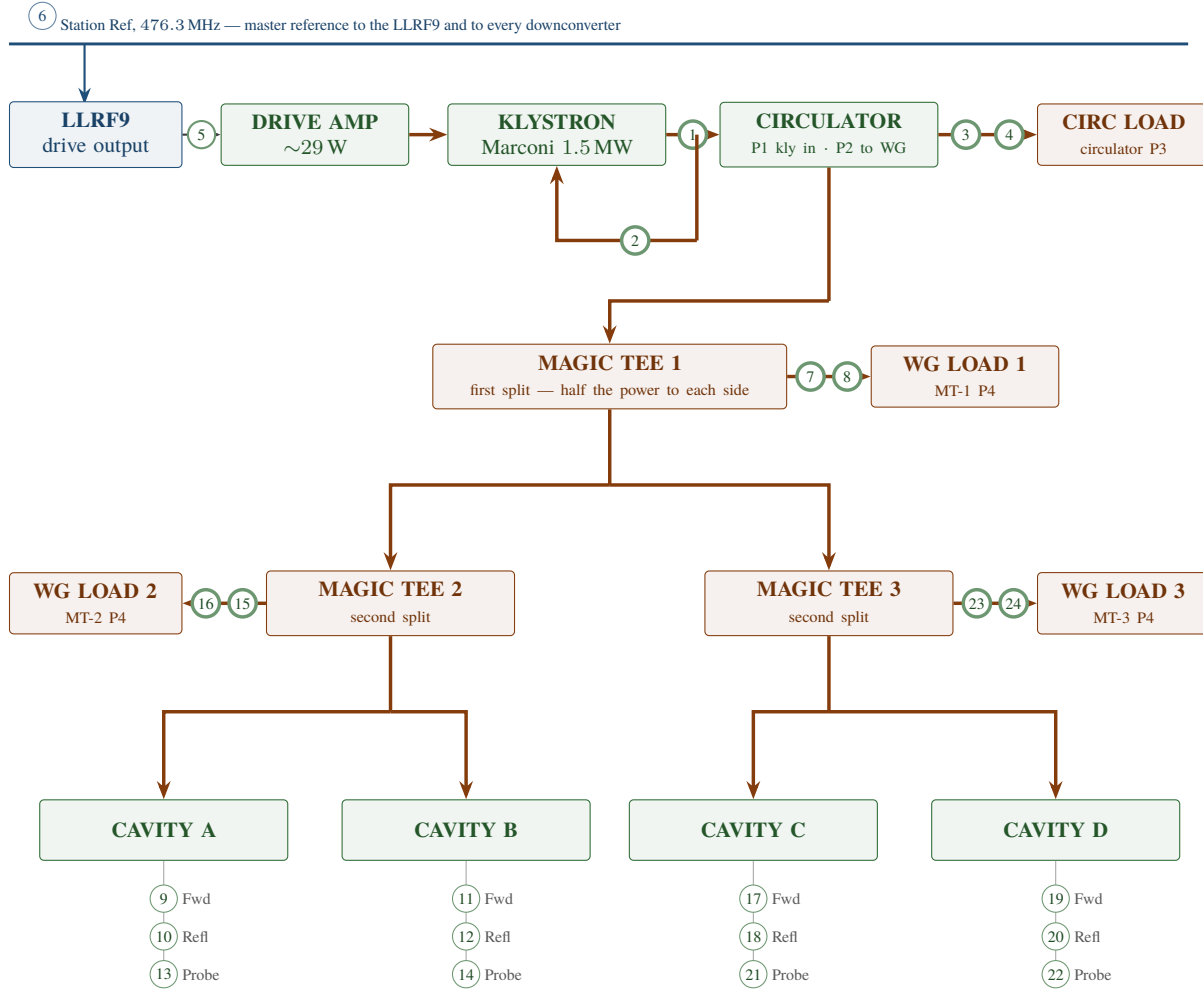


Figure 2: RF plant signal flow. Heavy lines are waveguide, light lines coaxial. Circled numbers are the directional-coupler channel indices used in Section 4.6.

5 Subsystem 1: LLRF Controller

Detailed reference: 11rf/11rf9

5.1 Legacy System

The legacy LLRF controller is a custom PEP-II analog RF Processor (RFP) module in a VXI chassis. It performs analog I/Q processing at $\sim \pm 90$ kHz bandwidth for the fast RF feedback loop. Associated modules include the Clock Module and three IQA (~ 1 MHz amplitude and phase detector) modules. The VXI chassis also hosts a processor running VxWorks RTOS with SNL (State Notation Language) control programs.

The legacy system processes 24 RF channels through the VXI system and uses analog-domain signal processing for feedback and calibration.

5.2 Upgraded System — Dimtel LLRF9/476

Two Dimtel LLRF9/476 units replace the entire VXI-based LLRF system (four units purchased; two active, two spares).

Hardware per unit:

- 3 × LLRF4.6 boards: each with Xilinx Spartan-6 FPGA (verify with Dmitry — may be Artix-7), 4 high-speed ADC channels, 2 DAC channels, 3 RF channels

- LO/Interconnect module: divide-and-mix LO synthesis for low phase noise, RF reference distribution, output amplification/filtering, interlock logic
- Linux SBC (mini-ITX): runs the built-in EPICS IOC (EPICS Base 3.14 — note: older than SSRL's EPICS7; migration plans TBD)
- Thermal stabilization: aluminum cold plate with 3 TEC modules under PID control, only available to board1&2
- Power supply: 90-264 VAC auto-ranging
- 3U 19" rack chassis

LO Frequency Plan (LLRF9/476) — nominal frequency 476.3 MHz (current: 476.3051755 MHz):

Signal	Ratio to f _{rf}	Nominal Frequency (MHz)
Reference (f _{rf})	1	476.3052
IF	1/12	39.6921
Local Oscillator	11/12	436.6131
ADC Clock	11/48	109.1533
DAC Clock	11/24	218.3065

Auxiliary I/O per unit:

- 8 slow ADC channels (12-bit, selectable ranges, galvanically isolated)
- 12 slow DAC outputs (AD5644)
- Opto-isolated interlock input (3.3V/5V/24V selectable — confirm exact voltage logic levels from test summaries)
- Interlock output (+4.75V high / <0.1V tripped, 220 Ω)
- Two opto-isolated trigger inputs

5.3 Two-Unit Configuration for SPEAR3

Unit 1 — Field Control & Tuner Loops:

Board	Ch0 (Ref)	Ch1	Ch2	Ch3	Output	Thermal Stabilization
BRD1	Station Ref	Cav A Probe	Cav B Probe	Cav A Fwd	Klystron Drive	Y
BRD2	Station Ref	Cav C Probe	Cav C Fwd	Cav B Fwd	(Spare / Monitor)	Y
BRD3	Station Ref	Circ Load Fwd	WG Load 2 Fwd	WG Load 3 Fwd	(Not used)	N

- **Feedback Architecture:** Cavity probes A and B can be monitored on BRD1 with proportional and integral control driving output 1. Probes C and D can be monitored on BRD2 with integral control driving output 2. An external power combiner combines the outputs to generate a signal containing an integral of all four cavities. The outputs of the two boards are synchronized via phase controls.
- **Tuner phase data:** All 4 cavity probe phases available at 10 Hz for tuner control

Unit 2 — Monitoring & Interlocks:

Board	Ch0 (Ref)	Ch1	Ch2	Ch3	Output	Thermal Stabilization
BRD1	Station Ref	Cav D Probe	Cav D Fwd	Drv Fwd	(Not used)	Y
BRD2	Station Ref	Cav D Refl	Kly Refl	Kly Fwd	(Not used)	Y
BRD3	Station Ref	Cav A Refl	Cav B Refl	Cav C Refl	(Not used)	N

- **Reflected power monitoring:** All 4 cavity reflected + klystron reflected for arc/mismatch detection
- **Interlock chain:** Reflected power events → Unit 2 interlock output → Interface Chassis → disables Unit 1 drive
- **No drive output required** from Unit 2

Remaining RF signals: The 6 signals not covered by the two LLRF9 units (Waveguide 1 Load Fwd, Circulator Load Refl, Station Reference, WG Load 1/2/3 Reflected) are monitored by the Waveform Buffer System — see Section 11.

5.4 Key Performance Specifications

Parameter	Value
Direct loop delay	270 ns
RF input channels	9 per unit (18 total)
RF input range	+2 dBm full-scale
ADC resolution	12-bit
Setpoint profile points	512
Setpoint step time range	70 μ s to 37 ms per step
Waveform samples/channel	16,384
Scalar readback rate	10 Hz
Phase readback resolution	Sub-degree
Interlock timestamp resolution	± 17.4 ns

5.5 Interfaces

- **RF inputs:** 50 Ω SMA from cavity probes, forward couplers, reflected couplers, station reference
- **RF output:** SMA to drive amplifier (Unit 1 only, thermally stabilized, BRD1 or BRD2)
- **Interlock I/O:** LEMO connectors; Unit 1 external interlock input from Interface Chassis, Unit 2 interlock output to Interface Chassis (confirm exact voltage logic levels from LLRF9 test summaries)
- **Slow ADC:** DA-15 connector; HVPS monitoring signals, auxiliary sensors
- **Ethernet:** Channel Access for EPICS communication (default ports 5064/5065)
- **Unit 1 $\leftrightarrow$ Unit 2:** Interlock daisy-chain (e.g. Unit 2 reflected power trip disables Unit 1 drive)

6 Subsystem 2: High Voltage Power Supply (HVPS)

Primary source references: Architecture: hvps/architecture/

6.1 Power Section (Retained)

Based on the proven PEP-II design architecture from 1997, this legacy system has been adapted for SPEAR3 operational requirements while maintaining the innovative star point controller topology and multi-layer arc protection system. The HVPS converts 12.47 kV RMS 3-phase AC to regulated DC high voltage for the klystron cathode. It is **rated –90 kV at 27 A (2.5 MW)** and typically operates at ≈ -72 to -75 kV at 19–22 A (≈ 1.4 – 1.6 MW DC), which supports the klystron's ≈ 800 kW typical RF output. The power section, including transformers, thyristor stacks, filter inductors, secondary rectifiers, crowbar, oil system, and all power cabling, is retained unchanged.

Parameter	Value
Maximum output voltage	–90 kV DC
Maximum output current	27 A
Maximum output power	2.5 MW
Nominal operating voltage	–74.4 kV
Nominal operating current	22.0 A
Nominal operating beam current	500 mA
Rectifier topology	12-pulse, thyristor phase-controlled (two 6-pulse bridges, B+ and B–)
Number of HVPSs	2 (SPEAR1 and SPEAR2) — see note below

Spare-supply posture. The two supplies are **not** in a warm-standby arrangement. One supply is energised and the other is **off**. Roles are exchanged during scheduled downtimes (typically one in April and one in August); either supply may be the active one. Both are fully operational.

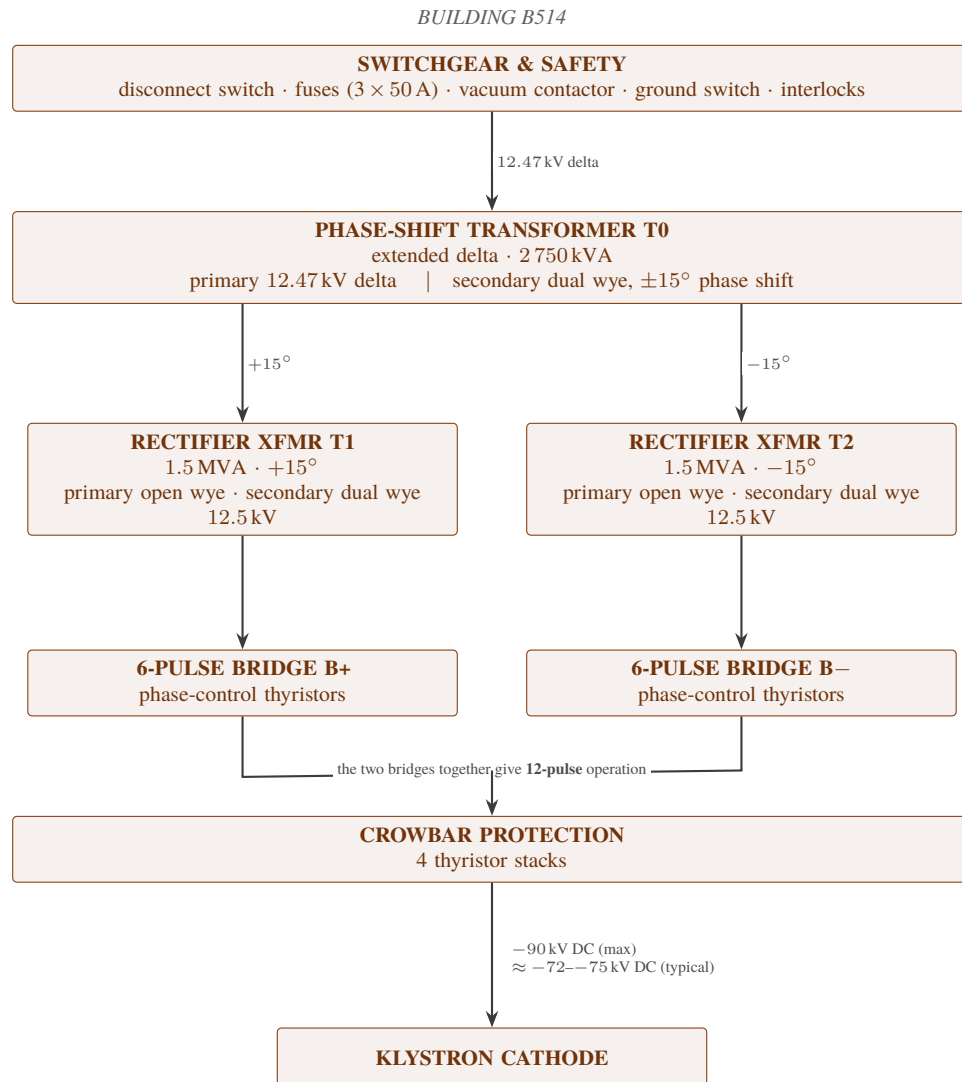


Figure 3: HVPS power chain in Building B514. The 15° phase offset between the two rectifier transformers is what turns two 6-pulse bridges into a single 12-pulse supply.

T0 rating — resolved September 2026. The phase-shifting transformer nameplate is **2750 kVA, 12.5 kV at 127 A RMS**, per NWL’s own electrical schematic EI-730-790-00-C0 (NWL # 39308), whose two figures are internally consistent ($2750 \text{ kVA} \div (\sqrt{3} \times 12.5 \text{ kV}) = 127 \text{ A}$). This supersedes the three figures previously in circulation: PS-341-360-01 §1.5 “350 kVA” (a decimal error — far too small to pass 2.5 MW), SD-730-790-01-C1 “3000 KVA” (a rounded label), and Sebek’s adopted 3.5 MVA estimate (made without access to the NWL drawing).

6.2 Legacy Controller

The legacy controller is housed in a Hoffman NEMA enclosure (34”×42”) in Building B118 and contains:

- **PLC:** Allen-Bradley SLC-500 (1747-L532 CPU) with I/O modules (OBSOLETE):

Slot	Module	Function
CPU	AB-1747-L532	Processor
1	AB-1747-DCM	Data Communications (Scanner)
2	AB-1746-IO8	8-pt Digital I/O — Ross switch coil (OUT3)
3	AB-1746-THERMC	Thermocouple inputs (SCR/air/transformer temps)
5	AB-1746-OX8	8-pt Relay Output — Contactor enable K4 (OUT2), Contactor On/Off (OUT1)
6	AB-1746-IB16	16 DC Input — PPS 1 (IN14), PPS 2 (IN15), Oil Level (IN8), Manual GRN SW (IN9)
7	AB-1746-IV16	16 DC Input (various permits)
8	AB-1746-NIO4V	4-ch Analog I/O — voltage setpoint output to Enerpro via regulator (N7:10 → OUT0)
9	AB-1746-NI4	4-ch Analog Input — Danfysik HVPS current (IN3)

Source: hvps/documentation/plc/plcNotesR1.docx (N7:10/N7:11 analysis, slot assignments)

- **Enerpro Firing Board:** FCOG6100 + FCOAUX60 (30° delayed triggering daughter board) — Current SCR gate driver
- **Regulator Card** (SD-237-230-14-C1): SLAC-designed analog voltage regulation loop with INA117 (difference amp), INA114 (instrumentation amp), OP77 (op-amp, 600 kHz GBW), BUF634 (high-current buffer), 4N32 optocoupler, and VTL5C opto-controlled variable resistor (OBSOLETE component)
- **Power Supplies:** SOLA $\pm 15V/+5V/24V$, Kepko 120V ($\times 2$), Kepko 5V/20A, Kepko 240V
- **PS Monitor Board** (SD-730-793-12): Monitors power supply health
- **Terminal Strips:** TS-5 (contactor, 15 terminals), TS-6 (grounding tank, 21 terminals), TS-3 (PPS LEDs), TS-7 (power distribution)
- **PPS Connector:** Originally Burndy circular 8-pin; possibly replaced with Souriau Trim Trio equivalent (GOB1208PNE) — exact current part number requires field verification

The SLC-500 PLC executes ladder logic for voltage regulation, contactor management, temperature monitoring, and — critically — PPS safety chain control:

- **PPS 1:** Controls K4 enable path for 12kV contactor; also has hardware fail-safe wired directly to OX8 relay input (bypasses PLC)
- **PPS 2:** Controls Ross grounding switch coil via Slot-2 IO8 OUT3 (Rung 0016) (Ross relay shorts out HV input to klystron.)
- **Hardware fail-safe:** K4 relay input side wired directly to PPS 1 signal (24 VDC), preventing PLC failure from keeping K4 energized without PPS enable

Sources: hvps/architecture/designNotes/HoffmanBoxPPSWiring.docx, hvps/architecture/designNotes/RFSysMPSRequirements.docx

Note on PPS chain status: The current system directly reports the status of one PPS chain and only indirectly reports the status of the other. The upgraded system should report the status of both via the use of an auxiliary contact on a relay energized by the PPS signal.

6.3 Upgraded Controller

The HVPS controller upgrade replaces the PLC and SCR gate driver while retaining the power section and Hoffman enclosure.

New hardware:

- **CompactLogix PLC:** Replaces SLC-500; handles voltage regulation, temperature monitoring, fault management, and EPICS communication
- **Enerpro FCOG1200 Board:** Upgraded SCR gate driver with modified RN4 resistors for SPEAR monitor winding impedance (2 M Ω)
- **Redesigned Analog Regulator:** Replaces obsolete regulator card components
- **Fiber Optic Interfaces:** Bidirectional communication with Interface Chassis:
 - STATUS: HVPS controller → HFBR-1412 transmitter → Interface Chassis (indicates control supply presence + no crowbar fired)
 - SCR ENABLE: Interface Chassis → HFBR-2412 receiver → HVPS controller (permit signal, active-high for fail-safe)
 - CROWBAR: Interface Chassis → HFBR-2412 receiver → HVPS controller (crowbar fire command)
 - Normal operation: All signals illuminated/active; signal loss = fault condition

Key change: The CompactLogix PLC is removed from the PPS safety chain. PPS functions (K4 relay, Ross switch control) are handled by a separate dedicated PPS interface box, not the Interface Chassis. The Interface Chassis handles only non-PPS interlocks. The PLC handles only non-safety functions: voltage regulation, temperature monitoring, and EPICS interface.

6.4 HVPS Controller Interface

With EPICS:

PV Category	Example PV	Update Rate
Setpoint	SRF1:HVPS:VOLT:CTRL.VAL	≤ 1 Hz (from Python)
Readback	SRF1:HVPS:VOLT:RBCK	~ 1 Hz
Status	SRF1:HVPS:STATUS:READY	~ 1 Hz
Control	SRF1:HVPS:CONTACTOR:CMD	On demand
Interlock	SRF1:HVPS:INTLK:SUMMARY	~ 1 Hz

Legacy HVPS Status PVs (for reference and troubleshooting):

Status Parameter	Function	Source
Contactors Closed	K4 relay status / Aux relay status	Slot-6 inputs
Over Voltage	Regulator card latch indication	SD-237-230-14-C1
Klystron Arc	Grounding tank + LHS trigger board detection	BNC-12 input
Transformer Arc	HVPS + RHS trigger board detection	BNC-0 input
Crowbar Status	Thyristor pulse detection	Enerpro board
PPS Chain Status	PPS 1 and PPS 2 availability (upgraded: both chains via relay aux contact)	Slot-6 IN14/IN15
SCR 1/2 Firing	Left/right side thyristor trigger status	Enerpro monitoring
AC Current	Input current measurement	Transformer monitoring
Temperature/Oil	SCR/air/transformer temperatures, oil level	Slot-3 thermocouples
12 kV Availability	Supply status/readiness	Switchgear feedback

Status Parameter	Function	Source
Fast Inhibit/Slow Start	Enerpro protection states	Gate driver board

Source: hvps/architecture/designNotes/rfedmHvpsLabelsPvs.docx (RF expert panel configuration)

With Interface Chassis:

- **Interface Chassis** (fiber optic): SCR ENABLE (in), CROWBAR inhibit (in), STATUS (out)
- **Crowbar Triggering Sources** (five independent sources for defense-in-depth protection):
 1. **SCR ENABLE** (fiber-optic): Interface Chassis → HVPS controller (loss of signal disables SCR triggers)
 2. **TRANSFORMER ARC TRIGGER** (BNC-0): Electronic signal from Hoffman box (AC current measurement or Stangenes transformer arc detection)
 3. **KLYSTRON CROWBAR** (fiber-optic): LLRF/Interface Chassis → HVPS controller (klystron protection command)
 4. **KLYSTRON ARC TRIGGER** (BNC-12): Electronic signal from Hoffman box (termination tank shunt sensing excessive current)
 5. **PLC FORCE CROWBAR** (Slot-5 OUT3): Active-low signal from SLC-500, monitored on BNC connector

Protection philosophy: Any single source can disable SCR triggers or fire crowbar; redundancy ensures protection even with single-point failures. Filter inductor stored energy is safely discharged through Enerpro logic and secondary rectifiers. Source: hvps/architecture/designNotes/RFSystemMPSRequirements.docx

- **EPICS Coordinator** (Ethernet): Voltage setpoint, readbacks, fault status
- **Waveform Buffer System:** HVPS voltage, current, inductor voltages monitored on 4 dedicated channels. HVPS voltage and current sent back to B132 as analog signals for RF MPS.
- **Switchgear:** Existing field cables to vacuum contactor controller and grounding tank

7 Subsystem 3: RF Machine Protection System (RF MPS)

Scope clarification: The RF MPS is the klystron/RF station equipment protection PLC. It is distinct from the facility-wide **SPEAR MPS**, which is an external system that provides a permit input signal to the RF MPS via the Interface Chassis.

7.1 Purpose and Protection Philosophy

The RF Machine Protection System (RF MPS) is the subsystem-level protection controller for the SPEAR3 RF station. Its purpose is to aggregate fault status from all RF subsystems and external safety systems (including the SPEAR MPS permit), manage the overall klystron/RF permit, and coordinate fault recovery.

The protection philosophy, as defined in the RF system MPS requirements (hvps/architecture/designNotes/RFSystemMPSRequirements.docx), is:

- Protect high-power elements (HVPS controller protects the HVPS; the RF MPS system protects the klystron and cavities) from damage by eliminating stored energy and disabling upstream power sources when faults occur.

- Redundant protection — multiple independent mechanisms protect each element (e.g., fiber-optic SCR disable, crowbar firing, contactor opening).
- Fail-safe design — all permits are active-high so that a broken cable or lost signal removes the permit.
- Graceful shutdown — when upstream elements trip, downstream elements are notified so they can shut down in an orderly fashion.

The RF MPS does **not** directly control the LLRF9 or HVPS hardware. Instead, it provides permit and control signals to the **Interface Chassis** (Section 8), which performs the fast hardware AND-logic and drives the actual interlock outputs to the LLRF9 and HVPS controller. This separation ensures that hardware interlock response times remain at the microsecond scale (Interface Chassis), while the RF MPS operates at PLC scan rates (~milliseconds) for fault aggregation, logging, and coordination.

7.2 Legacy System

The legacy RF MPS uses an **Allen-Bradley PLC-5** processor with **1771-series I/O modules**. It aggregates interlock signals from the RF system (via distributed analog modules and direct wiring) and the HVPS, and manages the RF permit chain. The SPEAR MPS beam permit and orbit interlock are **not** direct inputs to the legacy RF MPS PLC — they wire to the back connector of VXI Slot 5 (MPS Shutoff module) where they participate in the VXI-backplane RF permit logic. See Doc I §4.2.

In the legacy system, interlock coordination was distributed across analog modules, PLC I/O, and direct point-to-point wiring with no central coordination point. The PLC-5 communicated with the EPICS control system via a 1771-DCM scanner module.

Legacy RF MPS wiring diagrams: 33 sheets in `11rf/documentation/mpsWiringDiagrams/` (drawings `wd3403300200` through `wd3403303400`).

7.3 Upgraded System — ControlLogix 1756

The RF MPS PLC is upgraded from PLC-5 to **Allen-Bradley ControlLogix 1756** platform using a Rockwell Automation conversion kit. The ControlLogix platform provides modern Ethernet/IP communication, faster scan times, and long-term vendor support.

Status: Hardware assembled, software written, tested on SPEAR3 without RF power. Ready for EPICS IOC development and system integration.

7.4 RF MPS Outputs to Interface Chassis

In the upgraded system, the RF MPS communicates with the Interface Chassis via digital signals. These are the **only** direct hardware outputs from the RF MPS PLC to the interlock system:

Output Signal	Type	Description
RF MPS Summary Permit	Digital	Global RF permit from the RF MPS. One of several AND-gate inputs in the Interface Chassis. When removed, the Interface Chassis removes LLRF9 Enable and HVPS SCR ENABLE.
RF MPS Heart-beat	Digital	Watchdog signal. If the Interface Chassis stops receiving the heartbeat (indicating RF MPS PLC failure or communication loss), it removes all output permits.
RF MPS Reset	Digital	Clears all latched faults in the Interface Chassis first-fault register simultaneously. Required to restore system permits after a fault.

Important: The RF MPS does **not** directly drive the LLRF9 Enable, HVPS SCR ENABLE, or HVPS KLYSTRON CROWBAR signals. Those are Interface Chassis outputs, driven by the AND-logic of all input permits (see Section 8). The RF MPS contributes its Summary Permit as one input to that AND gate.

7.5 RF MPS Inputs from Interface Chassis

The RF MPS receives comprehensive status from the Interface Chassis via a multi-conductor digital cable:

Input Signal	Description
All input permit states	Status of every Interface Chassis input (LLRF9 Status, HVPS STATUS, SPEAR MPS permit, Orbit Interlock, Arc Detection, Waveform Buffer, expansion ports)
All output permit states	Status of every Interface Chassis output (LLRF9 Enable, HVPS SCR ENABLE, HVPS CROWBAR)
First-fault register	Identifies the initiating fault source when multiple faults cascade. Hardware-latched at the moment of initial fault detection.

This information allows the RF MPS to:

- Determine which subsystem caused a trip
- Log complete fault histories with timestamps
- Report detailed status to the EPICS coordinator for operator diagnostics

7.6 RF MPS Role in the Protection Chain

The RF system implements a layered protection architecture. The RF MPS operates at **Layer 3** (PLC scan rate, ~milliseconds):

Protection Layer	Subsystem	Re-sponse Time	Function
Layer 1	LLRF9 FPGA	<1 μ s	RF overvoltage interlocks, baseband window comparators, DAC zeroing, RF switch
Layer 2	Interface Chassis	<1 μ s	Hardware AND-logic, first-fault latching, HVPS fiber-optic control, LLRF9 enable
Layer 3	RF MPS PLC	~ms (PLC scan)	Fault aggregation, permit management, reset coordination, event logging
Layer 4	EPICS coordinator	~1 Hz	State machine, supervisory control, operator interface, auto-recovery

The RF MPS functions include:

- Aggregating fault status from all RF station subsystems (via Interface Chassis digital status lines)
- Receiving the SPEAR MPS permit as one of the AND-gate inputs via the Interface Chassis
- Providing the RF MPS Summary Permit to the Interface Chassis AND-gate
- Sending the heartbeat watchdog signal to the Interface Chassis
- Issuing the reset signal to the Interface Chassis to clear latched faults after a trip

- Logging fault events with timestamps for post-mortem analysis
- Providing redundant collector power calculation (from Waveform Buffer System data sent to RF MPS for independent verification)
- Reporting complete system status to the EPICS/MATLAB coordinator

7.7 EPICS Interface

The RF MPS ControlLogix PLC will provide an EPICS interface for operator monitoring and diagnostics. The PV namespace is SRF1:MPS:. Key PVs include:

PV	Type	Description
SRF1:MPS:PERMIT	BINARY	Overall MPS permit status
SRF1:MPS:FAULTS:ACTIVE	MULTI-BIT	Currently active fault sources
SRF1:MPS:FAULTS:FIRST	ENUM	First-fault identification (from Interface Chassis)
SRF1:MPS:FAULTS:COUNT	INTEGER	Accumulated fault event count

The EPICS coordinator reads these PVs to:

- Determine whether the MPS permit is active before allowing state transitions (e.g., OFF → STANDBY → ON)
- Force a transition to OFF state if MPS permit is lost during operation
- Display fault summaries and first-fault information to operators

7.8 Interfaces Summary

Interface Partner	Signal Direction	Signals	Medium
Interface Chassis	RF MPS → IC	Summary Permit, Heartbeat, Reset	Digital (multi-conductor cable)
Interface Chassis	IC → MPS	All input/output states, first-fault register	Digital (multi-conductor cable)
EPICS Coordinator	Bidirectional	Permit status, fault history, reset commands	EPICS Channel Access (Ethernet)

Note: External safety systems (SPEAR MPS permit, orbit interlock) connect to the **Interface Chassis** directly, not to the MPS PLC. The MPS receives their status indirectly via the Interface Chassis digital status lines. This design ensures that external safety permits participate in the fast hardware AND-logic ($<1 \mu\text{s}$) within the Interface Chassis, rather than being limited to PLC scan rates.

7.9 Source Documents

File	Description
hvps/architecture/designNotes/RFSystemMPSRequirements.docx	RF system machine protection requirements and protection philosophy (J. Sebek)
hvps/architecture/designNotes/interfacesBetweenRFSystemControllers.docx	Interface Chassis design — defines the MPS-to-Interface Chassis signal interface
llrf/documentation/mpsWiringDiagrams/	33 legacy MPS wiring diagram sheets (wd3403300200–wd3403303400)
pps/HoffmanBoxPPSWiring.docx	System-level architecture showing MPS in the upgraded system context
llrf/llrf9/iGp/dl_llrf/interlock_summary.edl	LLRF9 interlock summary display (EDM) — shows LLRF9-side interlock PVs that feed into the protection chain

8 Subsystem 4: Interface Chassis

Detailed reference: llrf/architecture/llrfInterfaceChassis.docx

8.1 Purpose

The Interface Chassis is a completely new subsystem that serves as the central hub for all hardware interlock coordination. It did not exist in the legacy system, where interlocks were distributed across analog modules, PLC I/O, and direct wiring with no central coordination point.

Key capabilities:

- **First-fault detection** — hardware latching identifies the initiating fault in cascade scenarios
- **Microsecond-scale response** — combinational logic with no processor in the critical path
- **Electrical isolation** — all external signals isolated via optocouplers and/or other galvanic isolation and fiber-optic transceivers
- **Fault latching** — all inputs latch when faulted until external MPS reset
- **Status reporting** — comprehensive digital reporting to MPS PLC

8.2 Interface Summary

Primary inputs: LLRF9 status, HVPS fiber-optic status, MPS permits/heartbeat/reset, SPEAR MPS, orbit interlock, arc detection, power monitoring, expansion ports

Primary outputs: LLRF9 enable, HVPS SCR ENABLE (fiber), HVPS CROWBAR (fiber), fault status reporting to MPS

Critical design consideration: The Interface Chassis creates a bidirectional feedback loop between LLRF9 and HVPS that requires careful recovery sequencing logic.

8.3 Implementation Status

Item	Status
Interface specification	In progress (J. Sebek, 11rf/architecture/11rfInterfaceChassis.docx)
Chassis design	Not started
PCB design	Not started
Fabrication	Not started

Note: The Interface Chassis is on the critical path for system integration.

9 Subsystem 5: Personnel Protection System (PPS) Interface

Detailed reference: pps/

9.1 PPS Overview

The PPS is the radiation and electrical safety interlock system managed by SLAC's radiation protection group. It controls two points in the HVPS power chain:

- **Chain 1:** Vacuum contactor (removes 12.47 kV input power to HVPS)
- **Chain 2:** Ross grounding switch (shorts the HVPS output to ground)

The PPS interface uses a GOB1208PNE (Burndy/Souriau Trim Trio) 8-pin circular connector mounted in a locked box on the Hoffman enclosure.

9.2 Legacy PPS Design (Problems)

In the legacy system, PPS signals are routed through the Hoffman Box:

- **PPS wiring exposure** — PPS wires terminate on terminal strips (TS-5, TS-6) inside the HVPS controller, co-located with non-PPS wiring
- **PLC in safety chain** — The Ross grounding switch is controlled by the SLC-500 PLC (Slot-2 IO8 OUT3, 120 VAC). If the PLC fails in a stuck-on state, the Ross switch coil stays energized (unsafe)
- **K4 relay via PLC** — Although the K4 relay coil uses PPS 1 voltage for its power rail (providing a hardware fail-safe), the enable path runs through PLC OX8 OUT2

9.3 Upgraded PPS Design

The upgraded design adopts the standard PPS interface solution already approved and in use by the PPS group for the 6575 modulator and gallery systems. It uses a dedicated PPS interface box.

PPS Interface Box Characteristics (per Ben Morris, March 5, 2026):

- Small Bud enclosure ($\approx 6 \times 5 \times 4$ ")
- Contains 4 relays, status LEDs (Permit A/B granted & enabled), and inhibit push-buttons
- Single board with one connector that the PPS group can lock with their collar
- Provides two independent permit channels + two readback channels (dry contacts)
- Completely isolated from the HVPS internals
- Labeled "PPS Interface – RSWCF required to open" (standard PPS practice)

PPS Signal Flow:

- **Chain 1 (Contactor):** PPS Enable 1 → PPS Interface Box → HVPS contactor relay direct drive → MX → L1 holding coil. S5 NC auxiliary contact readback via PPS Interface Box to PPS.

- **Chain 2 (Ross Switch):** PPS Enable 2 → PPS Interface Box → Ross switch direct drive. Ross NC auxiliary contact readback via PPS Interface Box to PPS.

Additional Safety Relays: Because the inhibit and safety-discharge wiring cannot be physically separated inside the existing HVPS and termination tank, a second relay is added in series with the inhibit/safety-discharge lines (one relay per channel). These relays are labeled “PPS Controlled – RSWCF required before work” with PPS-provided cable tags.

Key improvements:

- No PLC dependency for safety functions
- Uses proven PPS-approved design (identical to gallery systems)
- Electrical isolation of all PPS signals
- PPS wiring completely isolated from non-PPS equipment
- Clear visual feedback and independent inhibit capability
- Meets all current PPS requirements (two independent channels, lockable interface, visible verification)
- Eliminates the “unique RF situation” that the PPS group dislikes

Source: pps/pps_Ben.md (March 5 2026 meeting), pps/MSG\from\Jim\Sebek\to\Faya\about\PPS.md

9.4 PPS Regulatory Considerations

Any modification to PPS wiring, control logic, or readback paths requires:

- Review and approval by SLAC AD Safety Division
- Formal change control documentation
- Radiation safety verification testing
- Possibly a radiation safety work control form

This adds significant administrative scope beyond the engineering work and requires early engagement with the PPS/protection group.

10 Subsystem 6: Tuner Control System

10.1 Physical Mechanism

Each of the 4 RF cavities has a mechanical tuner that adjusts the cavity resonant frequency by moving a plunger into or out of the cavity volume. The tuner is driven by a stepper motor (Superior Electric M093-FC11 or equivalent) through a gear reduction mechanism.

Tuner parameters (from legacy documentation):

- Gear ratio: 80:1 (worm gear)
- Lead screw pitch: 10 turns/inch
- Microstep resolution: 0.002–0.003 mm per microstep (legacy)
- Linear potentiometer: Provides absolute position indication (not used in closed-loop control)
- Travel range: ~12 mm total mechanical range
- ON home positions: ~10.1–10.7 mm (cavity-dependent)
- Park positions: retracted position for safe storage

10.2 Legacy Controller

- **Motor controller:** Allen-Bradley 1746-HSTP1 (high-speed stepper module, OBSOLETE)
- **Motor driver:** Superior Electric SS2000MD4-M Slo-Syn PWM driver (OBSOLETE)
- **Control software:** SNL `rf_tuner_loop.st,v` — phase-based feedback with home reset, motion monitoring, and load angle offset computation
- **Phase source:** Legacy analog RFP module

10.3 Upgraded Controller

The tuner motor controller is being replaced with a modern motion controller. The leading candidate is the **Galil DMC-4143** (4-axis), which is supported by the LLRF9's built-in tuner control features and EPICS motor records. Other candidates under investigation include the Domenico/Dunning design and Danh's design.

Upgraded tuner control loop:

- LLRF9 measures cavity probe phase relative to station reference (10 Hz, sub-degree resolution)
- EPICS coordinator processes phase data and computes motor commands
- Motor commands sent via EPICS motor records to motion controller
- Motion controller drives stepper motors

LLRF9 tuner support: The LLRF9 includes built-in tuner control PVs (per cavity):

- `LLRF:TUNER:Cn:GAIN_P` — proportional gain
- `LLRF:TUNER:Cn:GAIN_I` — integral gain
- `LLRF:TUNER:Cn:OFFSET` — detuning phase offset
- `LLRF:TUNER:Cn:CLOSE` — loop state (open/closed)
- `LLRF:TUNER:Cn:MINFWD` — minimum forward power threshold

10.4 Load Angle Offset Loop

The load angle offset loop is a supervisory function that balances gap voltage across all 4 cavities by adjusting the individual tuner phase setpoints. In the legacy system this was embedded in `rf_tuner_loop.st,v`; in the upgrade it becomes a separate Python module (e.g., `load_angle_controller.py`) that:

- Reads all 4 cavity probe amplitudes from LLRF9
- Computes amplitude imbalance
- Adjusts individual tuner phase offset PVs to redistribute power

10.5 Tuner Interfaces

- **LLRF9 Unit 1:** Provides 10 Hz phase measurements for all 4 cavities (BRD1: Cav 1,2; BRD2: Cav 3,4)
- **Motion Controller:** EPICS motor records via Ethernet
- **Stepper Motors:** Power and encoder cables to tunnel (existing field cabling)
- **EPICS Coordinator:** Tuner manager module processes phase data, commands motor moves, manages load angle

10.6 Risk Note

The tuner motor controller is identified as the hardest-to-prove subsystem. Previous attempts with candidate controllers have had reliability issues. The mitigation plan is to test the chosen controller on the SPEAR3 booster tuners first before committing to storage ring installation.

11 Subsystem 7: Waveform Buffer System

Detailed reference: `llrf/architecture/WaveformBuffersforLLRFUpgrade.docx`

11.1 Purpose

The Waveform Buffer System is a new signal conditioning and monitoring chassis that extends the LLRF9's RF monitoring capabilities and adds dedicated HVPS signal monitoring. It serves three functions:

- **Extended RF signal monitoring** — 8 RF channels with circular waveform buffers for pre-fault capture; monitors the 6 RF signals not assigned to the two LLRF9 units
- **HVPS signal monitoring** — 4 channels (voltage, current, 2 inductor voltages) with circular buffers
- **Enhanced klystron collector power protection** — computes DC power minus RF power (vs. legacy forward power proxy) and triggers a hardware trip if collector power exceeds the limit

11.2 Channel Configuration

RF Channels (8):

The two LLRF9 units together cover 18 of the 24 monitored RF signals (see Section 5.3). The 6 remaining signals are routed to the Waveform Buffer RF inputs:

Channel	Sig- nal #	Signal	Purpose
1	7	Waveguide Load 1 Forward Power	Slow monitoring of WG Load 1 forward power; signal rarely exceeds 2 kW.
2	4	Circulator Load Reflected Power	Indicates circulator load match quality
3	6	Station Reference Power	Monitors 476 MHz reference level; loss of reference triggers fault. Note: Dmitry's LLRF9 module may also measure this.
4	8	Waveguide Load 1 Reflected Power	Monitors WG Load 1 match quality
5	16	Waveguide Load 2 Reflected Power	Monitors WG Load 2 match quality
6	24	Waveguide Load 3 Reflected Power	Monitors WG Load 3 match quality
7	—	(Spare — consider klystron forward power coupled version for collector power calculation)	Reserved
8	—	(Spare)	Reserved for future use

HVPS Channels (4):

Channel	Signal	Conditioning	Purpose
1	HVPS Voltage	Voltage divider	Regulation monitoring
2	HVPS Current	Current transformer	Load monitoring
3	Inductor 1 Voltage	Voltage divider	Firing circuit health
4	Inductor 2 Voltage	Voltage divider	Firing circuit health

11.3 Key Features

- **Circular buffers:** Continuous acquisition at kHz rate; freeze on fault trigger for pre/post-fault capture (~100 ms pre-fault data)
- **Analog comparator trips:** Hardware-based threshold detection on each channel; comparator outputs feed Interface Chassis
Enhanced collector power protection algorithm (improvement over legacy forward power proxy):

$$DC_Power \backslash = \backslash HVPS_Voltage \backslash \times \backslash HVPS_Current$$

$$RF_Power \backslash = \backslash Klystron_Forward_Power$$

$$Collector_Power \backslash = \backslash DC_Power \backslash - \backslash RF_Power$$

$$IF \ Collector_Power > Collector_Limit \ THEN \ Trip$$
This provides direct collector power calculation vs. the legacy system's forward power proxy method.
- **EPICS interface:** Waveform readout, threshold configuration, trip status

11.4 Interfaces

- **RF signal inputs:** RF detectors on cavity forward/reflected couplers (signal conditioned)
- **HVPS signal inputs:** Voltage dividers and current transformers from HVPS
- **Interface Chassis:** Comparator trip outputs (digital) feed into Interface Chassis permit logic
- **EPICS Coordinator:** Waveform readout, configuration, collector power trend monitoring

12 Subsystem 8: Arc Detection System

12.1 Purpose

The arc detection system is a new subsystem that provides optical monitoring of the RF cavity windows and klystron for electrical arcing. The legacy system had a non-functional arc detection capability that is being replaced entirely.

12.2 Technology

Microstep-MIS Waveguide Arc Detectors: Optical sensors that detect the light flash produced by an electrical arc inside a waveguide or cavity viewport. These are commercial off-the-shelf devices providing:

- Optical fiber sensors mounted at cavity window viewports, klystron window, and circulator
- Receiver units, each accepting up to 2 sensor fiber inputs with one independent dry-contact relay output per input channel
- Response time on the order of microseconds
- Configurable sensitivity thresholds

12.3 Installation

There are **10 sensors** total (updated from original design):

- 8 sensors mounted on cavity window viewports (two per cavity: Cav A, B, C, D)
- 1 sensor mounted on the klystron window
- 1 sensor monitoring the circulator

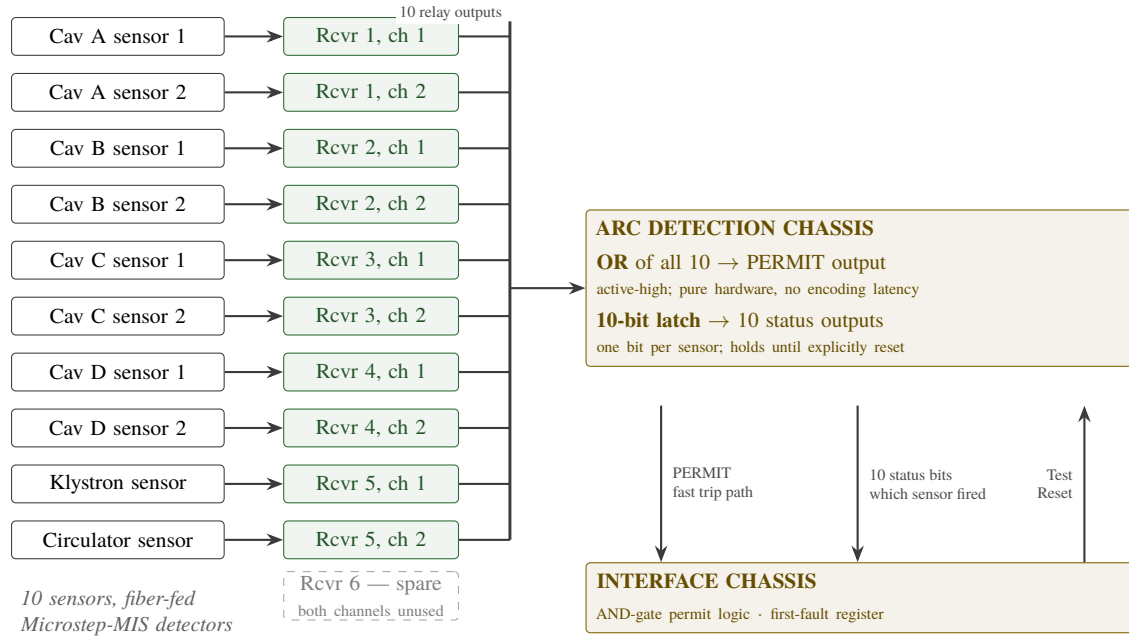


Figure 4: Arc detection signal path. The two outputs run in parallel: an all-hardware OR gives the fast trip, while the 10-bit latch preserves which sensor fired for post-mortem analysis.

Mechanical mounting requires custom adapters for the existing CF flange viewports (see 11rf/arcDetector/Mechanical/Reference/ for viewport specifications). Sensor fibers run to 6 Microstep-MIS receiver units located in B132 (5 active, 1 spare). Receiver-to-sensor assignment:

Receiver	Ch 1	Ch 2
Rcvr 1	Cav A sensor 1	Cav A sensor 2
Rcvr 2	Cav B sensor 1	Cav B sensor 2
Rcvr 3	Cav C sensor 1	Cav C sensor 2
Rcvr 4	Cav D sensor 1	Cav D sensor 2
Rcvr 5	Klystron sensor	Circulator sensor
Rcvr 6	(spare)	(spare)

12.4 Signal Path and Design

The Arc Detection Chassis routes both paths to the Interface Chassis — the fast trip permit on one wire, and the 6-bit fired-detector identification on six separate status lines. This centralizes all arc signal handling in the Interface Chassis, consistent with its role as the hub for all protection signals.

- **Fast trip path:** The 10 relay outputs (one per sensor, from 5 active receivers) are OR-ed in the Arc Detection Chassis into a single active-high permit signal. Any arc event immediately de-asserts it, entering the Interface Chassis AND-gate alongside all other permits. This path is purely hardware with no encoding latency.
- **Diagnostic identification path:** Each of the 10 relay outputs simultaneously sets an individual bit in a 10-bit latching register inside the Arc Detection Chassis. All 10 bits are wired as separate status inputs

to the Interface Chassis. The Interface Chassis includes these bits in its fault status word, which the RF MPS PLC reads to identify exactly which sensor fired. The latch holds state until explicitly reset, preserving identity through the fault response sequence.

Routing both paths to the Interface Chassis is cleaner than wiring the diagnostic path directly to the RF MPS: all arc-related signals are centralized in one place, no extra wiring runs to the MPS PLC are needed, and the Interface Chassis first-fault register can incorporate arc identification alongside all other fault sources.

12.5 Permit Logic

- The single arc permit output is **active-high** (permit present = no arc); any arc event drives it low — consistent with fail-safe convention throughout the Interface Chassis
- The Interface Chassis removes the RF permit immediately upon any arc event
- The 10 identification bits in the Interface Chassis fault status word indicate which individual sensor fired; the latch holds until reset
- If multiple sensors fire simultaneously, all corresponding bits are set
- Reset of the latch requires an explicit EPICS command via the RF MPS after the fault is acknowledged

12.6 Interfaces

- **Arc Detection Chassis** → **Interface Chassis**: 1 permit signal (OR of all 10 sensors, fail-safe active-high) + 10 latched status bits (one per sensor) + Test + Reset signals
- **Interface Chassis** → **RF MPS PLC**: Arc permit state and 10-bit fired-detector ID included in the Interface Chassis fault status word
- **RF MPS / EPICS**: Per-sensor trip status PVs, fired-detector identification, event count, latch reset command
- **EPICS Coordinator**: Reads arc fault status for operator displays and state machine fault handling

13 Subsystem 9: Klystron Cathode Heater

Detailed reference: <11rf/documentation/filamentHeater/sd3403110002.pdf>

13.1 Legacy System

The system was originally designed for the PEP-II B-Factory program at SLAC by P. Corredoura and the PEP-II LLRF group, and was subsequently adapted for use in the SPEAR3 storage ring RF system at SSRL. It has been in continuous service for over 25 years.

Key Design Characteristics:

Motor-driven variac providing continuously variable AC voltage

10:1 step-down toroidal isolation transformer (T1)

Solid-state relay (SSR) for on/off switching

Comprehensive monitoring: AC voltmeter, AC ammeter, current transformer (Texmate CT), and elapsed-time meter

Remote control interface via Allen-Bradley (A/B) PLC digital/analog I/O

Front-panel indicators (DS1, DS2 green LEDs) and local metering

Power Supply Capabilities:

Parameter	Legacy
AC Input	120 VAC, 60 Hz
Maximum Power Rating	~1 kW (1000 W)
Nominal Operating Power	~500 W (actual sustained operation)
Nominal Operating Voltage	68 V (AC, at transformer input)
Nominal Operating Current	7.3 A
Transformer Ratio	10:1 step-down
Secondary Output (Post-Transformer)	~6.8 V RMS at 73 A
Maximum Rating	14.0 V RMS @ 71 A

13.2 Upgraded System

The **Chroma programmable AC power supply** is the selected replacement. The unit is in hand, has been tested, and is ready to proceed. It is already in use elsewhere at the lab for the same heater application, providing a proven repair path and known operational characteristics.

Key advantages over the legacy variac:

- Analog voltage programming input — allows the RF MPS to directly ramp output voltage via a DAC signal, replacing the slow motor-driven variac
- Ethernet interface with VISA-type ASCII command set — enables EPICS monitoring and enable/disable control via an ASYN driver module
- Solid-state, no mechanical wear
- Known, in-lab support and repair path

Parameter	Legacy	Upgraded (Chroma)
Control method	Motor-driven variac	Programmable AC supply, analog voltage ramp
Response time	Seconds–minutes	<100 ms
Voltage regulation	$\pm 1\%$?	$\pm 0.1\%$
Reliability	Mechanical wear	Solid-state (>50,000 hr MTBF)
EPICS integration	Limited	ASYN module over Ethernet (monitoring + enable/disable)

Control architecture:

- **MPS controls the supply** via an analog ramp on the programming input. The RF MPS PLC DAC output drives the Chroma voltage programming pin, setting the output level. This keeps the safety-critical ramp control in hardware (MPS) with fast response.
- **EPICS communicates via Ethernet** for monitoring and enable/disable. The Chroma Ethernet interface supports VISA-type ASCII commands. An ASYN EPICS module will be developed to provide readback PVs (output voltage, current, status) and to send enable/disable commands from the EPICS coordinator.
- **Existing V/I metering retained:** The current analog voltage and current meters (feeding MPS analog inputs) are kept in place to provide hardware-level monitoring of the output within rating limits, independent of the Ethernet path.

Key design considerations:

- MPS ramps the Chroma output voltage via analog programming input for controlled warm-up and cooldown; no mechanical components

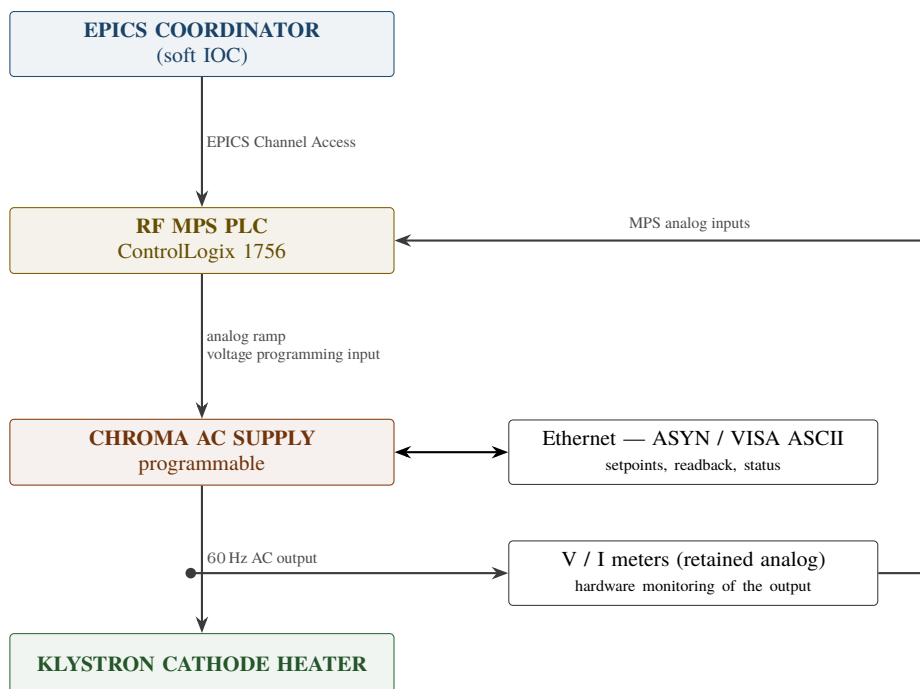


Figure 5: Klystron cathode heater controller architecture. Three paths are deliberately separate: the MPS ramps the Chroma output through the *analog* programming input, so controlled warm-up and cooldown do not depend on the network; the Ethernet link carries setpoints and readback only; and the existing analog V/I meters are retained, so the delivered output is monitored in hardware independently of both. There are no mechanical components anywhere in the control path.

- ASYN EPICS module required to interface to Chroma Ethernet (VISA ASCII command set); module to be developed
- Existing analog V/I metering feeds MPS hardware inputs directly — hardware protection path is independent of Ethernet/EPICS
- Automated warm-up, standby, and cooldown sequences managed by the EPICS coordinator
- Heater interlock: klystron does not receive an RF permit until the heater has reached operating power and the soak timer has elapsed; EPICS provides an expert override for short power dips

13.3 Interfaces

- **RF MPS → Chroma (hardware control):** MPS PLC DAC output drives the Chroma analog voltage programming input. Output is ramped up/down under MPS control for warm-up and cooldown sequences. MPS also provides the hardware enable/disable signal.
- **EPICS → Chroma (monitoring + soft control):** ASYN EPICS module communicates over Ethernet using VISA-type ASCII commands. Provides readback PVs for output voltage, current, and status; sends enable/disable commands from the EPICS coordinator.
- **Existing V/I metering (retained):** Legacy analog voltage and current meters feed MPS analog inputs directly, providing hardware-level protection independent of the Ethernet path. Output is monitored against rating limits.
- **HVPS coordination:** Heater must be at operating temperature before HVPS enable; HVPS must be off before heater cooldown.

- **RF MPS permit:** Heater “ready” status required for RF permit. Hardware heater current and voltage monitored by MPS analog inputs.

14 Subsystem 10: Control Software

Detailed reference: `spear-rf-code-legacy/rfApp/src/seq/, Docs_JS/LLRF0peration_jims.docx`

14.1 Legacy Software

The legacy control software consists of SNL (State Notation Language) programs running on VxWorks RTOS in the VXI crate processor:

Program	Lines	Function
<code>rf_states.st,v</code>	2,227	Master state machine (OFF/PARK/TUNE/ON_CW), turn-on, fault handling
<code>rf_dac_loop.st,v</code>	290	DAC control loop (gap voltage regulation)
<code>rf_hvps_loop.st,v</code>	343	HVPS control loop (drive power regulation)
<code>rf_tuner_loop.st,v</code>	555	Tuner control ($\times 4$ cavities, phase-based, load angle)
<code>rf_calib.st,v</code>	2,800+	Calibration (analog offset nulling, coefficient calibration, ~ 20 min)
<code>rf_msgs.st,v</code>	352	Message logging, HVPS faults, TAXI error detection

14.2 Upgraded Software — EPICS/MATLAB Coordinator

Note: Much of the new software functionality will already be in the LLRF9 internal EPICS IOC (written by Dmitry). Additional supervisory software will be in EPICS/Python/MATLAB as needed.

The upgrade replaces all SNL code with an EPICS/MATLAB coordinator application that operates as a supervisory control layer at ~ 1 Hz. It is NOT in the fast safety path.

14.3 Key Design Principles

- **Separation of concerns:** Each module has a single responsibility
- **Hardware abstraction:** All hardware interfaces go through the EPICS layer
- **Configuration-driven:** All operational parameters in configuration files
- **Fault tolerance:** Graceful degradation when subsystems are unavailable
- **Diagnostics:** Structured event logging, LLRF9 + waveform buffer readout, system performance metrics, and logging
- **Testability:** Mock interfaces for all hardware dependencies
- **Safety delegation:** Hardware safety handled by Interface Chassis; software handles sequencing and coordination only

15 Control Loop Architecture

This section maps the control loops in the system, identifying which hardware subsystem closes each loop, the loop bandwidth, and the upgrade path.

15.1 Fast RF Feedback (Direct Loop)

This is the fastest loop in the system and is entirely closed inside the LLRF9 FPGA. The EPICS coordinator does NOT participate.

15.2 HVPS Supervisory Loop (~1 Hz)

This loop trims the HVPS voltage setpoint to hold the requested operating point while respecting vacuum, gap-voltage and klystron-power limits. **The supervisory function is unchanged by the upgrade;** only the endpoints it reads and writes change.

The legacy implementation is `rf_hvps_loop.st,v` (343 lines), which runs a 0.5 s cycle with a 10 s hard deadline and operates in one of three states — off, proc and on:

- **PROC (cavity processing):** each cycle the loop compares klystron forward power against its limit and checks the cavity vacuum and gap-voltage alarm severities. If any is bad it applies a fixed negative step to the voltage setpoint; if all are good it applies a fixed positive step. This is what walks the cavities up in voltage during conditioning.
- **ON (normal operation):** the correction step is taken from the drive-power error when the direct loop is closed, or from the station gap-voltage error when it is not. The result is clamped to the configured voltage minimum and maximum.
- **Tolerance supervision:** if the voltage readback disagrees with the requested value by more than the allowed difference for 10 consecutive cycles, the loop holds the setpoint and raises a status code rather than continuing to drive. A separate 10-cycle counter guards the case where the cavities are already at their voltage limit.

The loop reports one of 16 status codes (for example GOOD, CAVV_LIM, VOLT_TOL, VOLT_LIM, VACM_BAD) together with a human-readable string, so the operator sees *why* the loop is holding rather than only that it is.

In the upgrade the same algorithm runs in the EPICS coordinator at ~1 Hz. The differences are all at the interfaces: the voltage setpoint and readback move over Ethernet to the CompactLogix HVPS PLC instead of through the Allen-Bradley serial link to the SLC-500, and the gap-voltage and drive-power inputs come from the LLRF9 units rather than from the RFP and IQA modules. Because the Waveform Buffer System now measures HVPS voltage and current directly, the loop also has a true power reading available where the legacy loop had only a forward-power proxy.

15.3 Tuner Control Loops ($\times 4$, ~1 Hz)

One loop per cavity nulls the cavity detuning by driving the tuner stepper motor until the probe-to-forward phase difference reaches its setpoint. **The control law is unchanged by the upgrade.**

The legacy implementation is `rf_tuner_loop.st,v` (555 lines), instantiated four times — once per cavity — with five states (`loop_init`, `loop_unknown`, `loop_off`, `loop_reset`, `loop_on`). Each cycle it:

1. reads the cavity phase error (probe phase minus forward phase);
2. converts it to a tuner position delta through the per-cavity gain;
3. clamps the resulting position to the motor drive limits and skips the move entirely if the change falls inside the deadband, so the tuner does not chatter;
4. commands the move and then waits for the motor's done-moving flag before trusting the new position.

Guards are on the motion rather than the arithmetic: five consecutive failed moves raise a motor-stall status, a missing or bad-severity phase measurement stops motion rather than moving on stale data, and forward power below the configured minimum disables the loop. There is a separate homing sequence with distinct home positions for the ON and PARK states, retried up to five times. The loop reports 14 status codes distinguishing the warning cases (at a drive limit, not at requested position, taking too long to move) from the non-functional cases (bad tuner readback, missing measurements, RF power too low).

In the upgrade the phase measurement comes from the LLRF9 units, which deliver all four cavity probe phases at 10 Hz with sub-degree resolution — an order of magnitude better than the legacy IQA chain — and the motor command goes to the Galil DMC-4143 through EPICS motor records instead of to the Allen-Bradley 1746-HSTP1. The LLRF9 also provides its own built-in tuner control PVs (GAIN_P, GAIN_I, OFFSET, CLOSE, MINFWD per cavity), so the loop can either be closed inside the LLRF9 or run in the coordinator using LLRF9 phase data. Which of the two is used is a commissioning decision; the deadband, drive-limit and stall guards above are required either way.

15.4 Load Angle Offset Loop (~1 Hz)

This outer loop balances gap voltage across the four cavities by biasing the individual tuner phase setpoints, as described in Section 10.4. It reads the four cavity probe amplitudes, computes the imbalance, and adjusts each cavity's phase offset to redistribute power.

Legacy status: the framework exists inside `rf_tuner_loop.st,v` and the offset is computed in the EPICS database, but it is *not used* on SPEAR3 — the phase offset fed to the tuner servo is fixed at zero, so the legacy tuners run in pure resonance-seeking mode. The mechanism is inherited from PEP-II, where it regulated cavity voltage in park mode.

In the upgrade the loop becomes a distinct supervisory module in the EPICS coordinator rather than logic buried in the tuner sequencer, writing to the per-cavity tuner phase offset. Whether it is enabled is an operational decision; if it is, it needs an integrator limit and a hold condition tied to tuner loop health, because an outer loop writing into a servo that is itself faulted is a known way to walk a tuner into a limit.

15.5 Station State Machine

The station sequencer owns turn-on, turn-off, mode changes and the response to a fault. **The sequencing responsibility is unchanged by the upgrade;** the state set is simplified because much of the legacy complexity existed to manage PEP-II loops that SPEAR3 does not use.

The legacy implementation is `rf_states.st,v` (2,227 lines) — by a wide margin the largest piece of legacy control software. It comprises 23 states across three concurrent state sets:

- **5 operating states:** OFF, PARK, TUNE, ON_FM, ON_CW.
- **12 transition substates** that carry the station between them, including the orderly shutdown, the station-reset retry, and the fault-file capture.
- **5 loop-protection substates** that ramp gap voltage up and down and ramp the direct and comb loop gains, plus a concurrent fault-file writer.

Not every transition is legal. PARK cannot go directly to either ON state — it must pass through TUNE — and returning from ON_CW to a lower state is routed through a controlled fallback rather than a direct jump. On a fault the sequencer captures the fault waveform files, performs an orderly shutdown, and then attempts an automatic restart up to a retry limit; a forced-fault latch blocks that automatic restart until an operator clears it.

In the upgrade the sequencer moves to the EPICS coordinator and the state set collapses, because ON_FM and the comb and gap-voltage-feedback ramps it existed to manage are PEP-II features that SPEAR3 does not use (Section 15.7). Ramping drive power and closing the field loop become LLRF9 operations;

the HVPS turn-on and turn-off sequences become CompactLogix operations; tuner homing becomes a Galil operation. What the coordinator retains is exactly the part that was always supervisory: deciding which transitions are legal, ordering the subsystem operations, arming and clearing the fault response, and enforcing the retry limit and the forced-fault latch.

The one behaviour that must be preserved carefully is the interaction with the protection chain. The legacy sequencer could afford to be in the trip path because it ran on a real-time kernel; the upgraded coordinator is explicitly *not* in the fast safety path (Section 17.1), so the state machine must react to a trip that the Interface Chassis has already executed, rather than being the thing that executes it.

15.6 Calibration

Calibration nulls the offsets through the RF signal chain and measures the modulator response so that commanded I/Q corresponds to actual field. **The need is unchanged by the upgrade; the implementation is entirely replaced.**

The legacy implementation is `rf_calib.st,v`, roughly 2,800 lines running a 28-state sequence that takes about 20 minutes. Each stage performs a binary search on a DAC channel, settling after each step, averaging a long I/Q record, and halving the step until the residual is within one LSB. It works through the cavity modulator terms, the direct loop, the comb loop, the summing and gain stages, the klystron stage and demodulator, and the final modulator null — on the order of sixty DAC channels. It aborts cleanly on operator interrupt, on a missing comb module, and when RF is not enabled, restoring the saved DAC values on the way out.

In the upgrade almost all of this disappears with the hardware it calibrated. The offsets being nulled were those of the analog RFP signal chain; the LLRF9 performs its own internal calibration of its ADC and DAC chains and provides it as a built-in function. What remains at the system level is the *station* calibration that no single unit can do for itself: establishing the phase and amplitude relationship between the two LLRF9 units, the cable-length and coupler-calibration constants that convert LLRF9 readings into cavity gap voltage and waveguide power, and the tuner phase reference per cavity. These are coordinator-level procedures operating on LLRF9 measurements rather than DAC-level nulling, and they should be expected to take minutes rather than the legacy twenty.

15.7 Eliminated Loops

The following legacy loops are eliminated in the upgrade because the LLRF9's digital feedback inherently provides their function (note: comb filter and GVF were used for PEP-II, not SPEAR3):

- **Ripple rejection loop** — LLRF9 digital feedback inherently rejects power-line ripple
- **Comb filter loop (CFM)** — used for PEP-II multi-bunch stabilization, not applicable to SPEAR3
- **Gap voltage feedback (GVF)** — used for PEP-II, not SPEAR3; cavity field stabilization now handled by LLRF9 vector sum feedback
- **4-way branching (DAC loop)** — Eliminated; LLRF9 controls all via single vector sum

16 Inter-Subsystem Interface Matrix

This matrix summarizes all physical and logical interfaces between subsystems in the upgraded system.

From To	LLRF9 #1	LLRF9 #2	HVPS PLC	RF MPS PLC	Interface Chassis	Wave- form Buffer	Arc De- tect	Motor Ctrl	Heater Ctrl	EPICS Coord
LLRF9 #1	—	Inter- lock daisy (LEMO)	—	—	Status (5V digital)	—	—	—	—	EPICS (Ether- net)
LLRF9 #2	Inter- lock daisy (LEMO)	—	—	—	Status (5V digital)	—	—	—	—	EPICS (Ether- net)
HVPS PLC	—	—	—	—	STATUS (fiber), SCR EN (fiber), CROWBAR (fiber)	—	—	—	—	EPICS (Ether- net)
RF MPS PLC	—	—	—	—	Summary, Heartbeat, Reset (digi- tal)	—	—	—	—	EPICS (Ether- net)
Inter- face Chas- sis	Enable (3.2V)	Enable (3.2V)	SCR EN, CROW- BAR (fiber)	Fault status (digi- tal)	—	—	—	—	—	—
Wave- form Buffer	—	—	—	—	Comparator trips (digi- tal)	—	—	—	—	EPICS (Ether- net)
Arc Detect	—	—	—	—	Permit + Latching	—	—	—	—	—
Motor Ctrl	—	—	—	—	—	—	—	—	—	EPICS (Ether- net)
Heater Ctrl	—	—	—	—	—	—	—	—	—	EPICS (Ether- net)
EPICS Coord	EPICS (Ether- net)	EPICS (Ether- net)	EPICS (Ether- net)	EPICS (Ether- net)	EPICS	EPICS (Ether- net)	—	EPICS (Ether- net)	EPICS (Ether- net)	—
Exter- nal (SPEAR MPS, Orbit)	—	—	—	—	24V permits	—	—	—	—	—

16.1 Interface Signal Types Summary

The matrix above uses six physically distinct interface types. The division matters because it is what makes the "no safety function depends on the network" claim in Section 18.2 true by construction.

Type	Used for	Between	Why this medium
Fiber optic	SCR ENABLE, CROWBAR, STATUS	Interface Chassis ↔ HVPS	Full galvanic isolation across a building boundary at high voltage; immune to the ground transients a crowbar event produces
Digital, opto-isolated	Permits, heartbeat, reset, fault status	Interface Chassis ↔ MPS, Waveform Buffer, Arc Detect	Deterministic, no protocol stack; response set by propagation not by scan rate
LEMO interlock	Interlock daisy-chain	LLRF9 #1 ↔ LLRF9 #2	Vendor-native interface; propagates a reflected-power trip between units without going through the Interface Chassis
Dry contact / solid-state relay	Arc detector outputs	Arc receivers → Arc Detection Chassis	Native output of the Microstep-MIS receivers; contact closure needs no supply on the sensor side
Analog	Heater voltage programming; HVPS voltage and current; V/I metering	MPS → Chroma; HVPS → Waveform Buffer and MPS	Continuous quantity with a hardware path that stays valid if the network is down
EPICS Channel Access	Setpoints, readbacks, waveforms, status, commands	EPICS coordinator ↔ all intelligent devices	Supervisory only; every device on it holds its last safe state if the link drops

Two conventions apply across all of them. Every permit is **active-high**, so a broken cable, a lost supply or a dark fiber removes the permit rather than granting it. And every interlock signal that crosses a chassis boundary is isolated — by optocoupler, by fiber, or by relay contact — so that no fault current or ground offset propagates between subsystems.

16.2 Open Interface Questions

The following interface details are not yet settled and are tracked here rather than being buried in the sections above. Each must be closed before the corresponding subsystem can be integrated.

Interface	Open question	Needed before
LLRF9 ↔ Interface Chassis	Exact voltage logic levels for the interlock input and output. The input is selectable (3.3/5/24 V) and the output is specified as 4.75 V high through 220 Ω; the Interface Chassis input and output stages must be designed to match.	Interface Chassis PCB design
EPICS version	The LLRF9 internal IOC runs EPICS Base 3.14, older than the SSRL site standard. Whether the units are migrated or bridged by a gateway is undecided.	Coordinator software design
LLRF9 FPGA	Board generation (Spartan-6 vs. Artix-7) to be confirmed with the vendor; affects available firmware features.	Firmware feature commitments
Network addressing	IP addresses and subnet allocation for both LLRF9 units, both PLCs, the motion controller and the Waveform Buffer are all TBD.	Any integration testing
Soft IOC location	Whether the coordinator IOC is hosted in B132 or B137 is not yet confirmed with the controls group.	Rack and network planning
Heater control	The ASYN/VISA EPICS module for the Chroma supply does not exist yet; the ASCII command subset to be supported is not defined.	Heater subsystem commissioning
Tuner control split	Whether the tuner loop is closed inside the LLRF9 or in the coordinator using LLRF9 phase data (Section 15.3).	Motion controller acceptance
PPS connector	The installed connector part number requires field verification; the original Burndy circular 8-pin may already have been replaced by the Souriau Trim Trio equivalent.	PPS Interface Box fabrication
Waveform Buffer spares	Two RF channels are unassigned. One is a candidate for a coupled klystron forward power signal feeding the collector power calculation.	Waveform Buffer PCB layout

17 Protection Chain and Interlock Architecture

17.1 Protection Philosophy

The upgraded system implements a layered protection architecture:

- **Layer 1 — LLRF9 FPGA interlocks** (<1 μs): RF overvoltage, baseband window comparators, internal housekeeping. These are the fastest protection and act within the LLRF9 itself to disable the DAC output.
- **Layer 2 — Interface Chassis hardware** (<1 μs from input change): Aggregates all permit signals and coordinates system-wide protection. All signals are optocoupler-isolated or fiber-optic. First-fault detection identifies the initiating event.
- **Layer 3 — RF MPS PLC** (~ms): Aggregates RF interlock status with external safety systems (SPEAR MPS, orbit interlock). Provides reset signal to Interface Chassis.

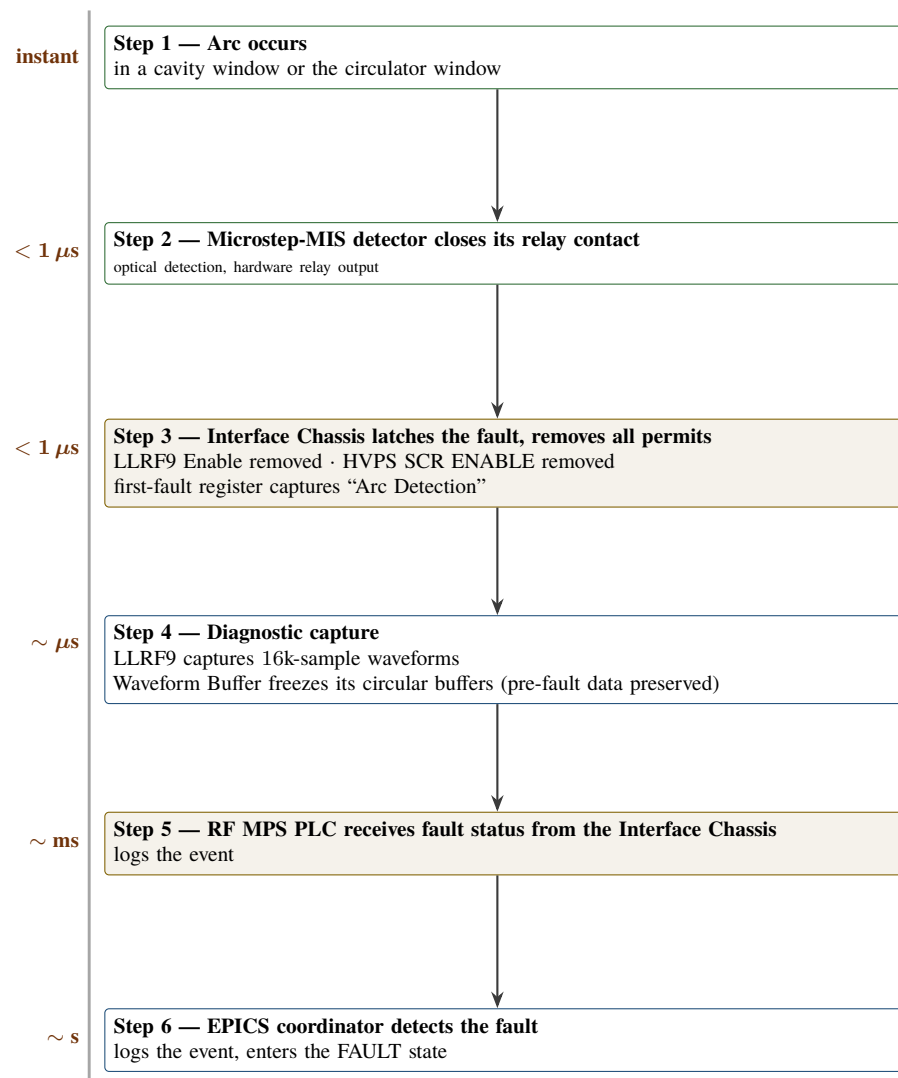


Figure 6: Fault propagation for an RF arc event. Each step is roughly three orders of magnitude slower than the one above it. Steps 1–3 are pure hardware: protection is complete before any programmable device is involved. Steps 4–6 are diagnostic and supervisory only — no protective action depends on them, so a PLC or IOC failure degrades observability but never safety.

- **Layer 4 — EPICS Coordinator** (~1 s): Supervisory monitoring, logging, fault analysis, and recovery sequencing. NOT in the fast safety path.

17.2 Fault Propagation Example — RF Arc Event

17.3 Fail-Safe Design

All critical subsystems are designed to fail safe:

Subsystem	Failure Mode	Safe State
Interface Chassis power loss	All outputs de-energize	LLRF9 disabled, HVPS SCR disabled (K4/Ross controlled by separate PPS interface box)
LLRF9 power loss	Interlock status goes low	Interface Chassis removes permits
HVPS PLC failure	STATUS signal lost	Interface Chassis removes permits
RF MPS PLC failure	Heartbeat lost	Interface Chassis removes permits
EPICS coordinator crash	No effect on safety	Hardware protection continues; supervisory control paused
Ethernet failure	No effect on safety	Hardware protection continues; supervisory control paused

18 Communication Architecture

18.1 EPICS Channel Access Network

All supervisory communication in the upgraded system uses EPICS Channel Access over Ethernet:

Device	IP Address	IOC Type	Key PV Prefix	Update Rate
LLRF9 Unit 1	TBD	Built-in Linux IOC	LLRF1:	10 Hz (scalars)
LLRF9 Unit 2	TBD	Built-in Linux IOC	LLRF2:	10 Hz (scalars)
HVPS CompactLogix	TBD	External EPICS gateway	SRF1:HVPS:	~1 Hz
RF MPS ControlLogix	TBD	External EPICS gateway	SRF1:MPS:	~1 Hz
Motion Controller	TBD	EPICS motor record IOC	SRF1:MTR:	On demand
Waveform Buffer	TBD	Dedicated IOC	SRF1:WFBUF:	~1 Hz / event
EPICS/MATLAB Coordinator	TBD	caproto / PyEPICS / MATLAB client	Various	~1 Hz

18.2 Non-EPICS Communication Paths

Path	Type	Purpose
Interface Chassis ↔ all hardware	Hardwired digital/fiber	Safety interlocks (no network dependency)
LLRF9 Unit 1 ↔ Unit 2	LEMO interlock daisy-chain	Fast interlock propagation
PPS ↔ PPS Interface Box	Hardwired (GOB1208PNE)	Personnel safety (no network dependency)

Path	Type	Purpose
Arc Detection → Interface Chassis	Dry contact relay and/or semiconductor switches	Arc interlock (no network dependency)

Critical design principle: All safety-critical communication paths are hardwired. No safety function depends on the Ethernet network or EPICS Channel Access. Network loss pauses supervisory control but does not compromise protection.

19 Implementation Phases and Risk Summary

19.1 Implementation Phases

Critical Constraint: The SPEAR3 LLRF upgrade has extremely limited flexibility for integration testing since bringing systems online directly impacts SPEAR operations. This severely constrains the available testing window and requires a fundamentally different approach than typical development projects.

Key Implications:

- Maximum standalone subsystem testing must occur before installation
- Integration testing must be incremental and carefully planned
- Some subsystems can be brought online for limited testing without full operational impact
- Full system integration testing window is minimal

19.1.1 Phase 1: Maximum Standalone Development

- **Software Framework:** Develop with simulated interfaces where live hardware unavailable
- **Tuner Testing:** Test Galil controller with Booster RF cavity
- **LLRF9 Installation:** Install both units, connect RF signals, and bring them online
- **RF MPS Online:** Bring RF MPS online for EPICS interface development
- **HVPS PLC Online:** Enables EPICS software development
- **Waveform Buffer Assembly:** Complete assembly and standalone testing
- **Interface Chassis Design/Fab:** Complete design and fabrication
- **HVPS Controller:** Design, fabrication, and installation
- **Arc Detector:** Design and fabrication
- **PPS Interface Box:** Design, fabrication, and commissioning
- **Test Stand 18:** Upgrade to work with the upgraded HVPS controller

19.1.2 Phase 2: Incremental Subsystem Integration

- **Tuner Test with SPEAR Cavity**
- **HVPS Integration:** Integrate HVPS PLC with EPICS coordinator (test at test stand T18)
- **RF MPS Integration:** Integrate RF MPS with LLRF9, waveform buffer, arc detector, HVPS PLC, interface chassis, and software coordinator with existing RF signals

19.1.3 Phase 3: Critical Path Integration

- **LLRF9 Integration — Basic RF Processing:** Verify vector sum, feedback loops, online calibration
- **EPICS Integration:** Configure IOC, verify all PV interfaces
- **End-to-End Interlock Testing:** Verify complete interlock chain (limited RF power)

19.1.4 Phase 4: Full Power Commissioning

- **Incremental Power Ramp:** Gradually increase power while monitoring all systems
- **Performance Validation:** Verify all success criteria are met
- **Operator Training:** Train operators on new system
- **Documentation:** Complete commissioning report

19.2 Status

Item	Status	Notes
LLRF9 (4 units)	Complete	—
RF MPS PLC modules	Complete	—
HVPS PLC modules	Complete (HVPS1, HVPS2, B44 Test Stand)	—
Enerpro FCOG1200 boards	Needed	5 boards
Arc detection (Microstep-MIS)	Needed	10 sensors and 5 process + 1 sensor & 1 process for spare
Waveform Buffer System	Needed	Designed
Interface Chassis	Needed	—
Heater Controller (programmable AC supply)	Needed	—
PPS Interface Box	Needed	—
Software	Needed	—

19.3 Key Technical Risks

Risk	Sever-ity	Status	Mitigation
Tuner motor con- troller reliability	High	In progress	Test on booster tuners first; evaluate multiple candidates
HVPS PLC code migration (SLC-500 → CompactLogix)	High	In progress	Reverse-engineer legacy code; build Test Stand 18 (B44)
PPS Interface ap- proval delays	High	In progress	Engage protection managers early; provide complete docu- mentation

Risk	Severity	Status	Mitigation
Interface Chassis logic design (LLRF9/HVPS feedback loop)	High	Not started	Design and simulate before fabrication; careful sequencing logic
Waveform Buffer System (new custom hardware)	Medium	Not started	Staged development: PCB design → assembly → testing → integration
EPICS/MATLAB Coordinator software	Critical	Not started	Largest untouched software scope: state machine, fault handling, auto-recovery sequences, operator interface, and integration with all subsystem IOCs. Must be developed in parallel with hardware integration.
RF MPS PLC software integration	High	Not started	MPS hardware is assembled and tested standalone, but EPICS IOC development and integration with Interface Chassis fault status, first-fault logic, and reset sequencing has not begun.
Arc Detection Chassis firmware/logic	Low	Not started	OR-gate and 6-bit latch logic. Straightforward design; prioritize as part of Interface Chassis design phase.
Communication latency (Ethernet/EPICS for ~1 Hz control)	Low	Validated	Proven in LLRF9 prototype commissioning

19.4 Success Criteria

Metric	Legacy Performance	Target
Amplitude stability	<0.1%	Same or better
Phase stability	<0.1 deg	Same or better
Tuner resolution	~0.002–0.003 mm/microstep	Improved (up to 256 microsteps/step)
Control loop response	~1 second	Same or better
Uptime	>99%?	Same or better
Fault diagnostics	Limited fault file capture	16k-sample waveform + circular buffer + first-fault

20 Appendix: Source Document Index

Every path below has been checked to exist. Note that the legacy SNL source is RCS-archived under `spear-rf-code-legacy/rfApp/src/seq/` with `,v` suffixes — there is no `llrf/legacyLLRF/` directory.

20.1 Governing project document

Path	Content
Designs/docx/SPEAR3_LLRF_PDR_R2.docx	SPEAR3 LLRF Upgrade Preliminary Design Review, R2 (28 April 2026) — the authoritative upgrade specification. Sections 1–19 of this report track it directly
Designs/docx/SPEAR3_LLRF_Upgrade_Project_Schedule.xlsx	Project schedule

20.2 LLRF Source Material

Path	Content
../spear-rf-code-legacy/rfApp/src/seq/rf_states.st,v	Legacy master state machine (23 states)
../spear-rf-code-legacy/rfApp/src/seq/rf_dac_loop.st,v	Legacy DAC control loop
../spear-rf-code-legacy/rfApp/src/seq/rf_hvps_loop.st,v	Legacy HVPS control loop
../spear-rf-code-legacy/rfApp/src/seq/rf_tuner_loop.st,v	Legacy tuner control loop
../spear-rf-code-legacy/rfApp/src/seq/rf_calib.st,v	Legacy calibration system
../spear-rf-code-legacy/rfApp/src/seq/rf_msgs.st,v	Legacy message logging
../spear-rf-code-legacy/codeReviewTechnicalNotes/	Nine-part code review of the legacy application
11rf/architecture/11rfInterfaceChassis.docx	Interface Chassis specification
11rf/architecture/WaveformBuffersforLLRFUpgrade.docx	Waveform Buffer System design
11rf/architecture/arcDetectorHardwareOptions.docx	Arc detection hardware selection
11rf/architecture/rfPowerDetector.docx	RF power detector selection
11rf/architecture/analogDesignComponents.docx	Analog component selection
11rf/documentation/LLRFOperation_jims.docx	SPEAR3 RF Station Operation Guide (J. Sebek)
11rf/documentation/LLRFUpgradeTaskListRev3.docx	Full project scope, procurement, costs
11rf/documentation/fiberOpticCableSignalControlRev3.docx	Fibre-optic signal and control routing
../11rf/tuners/galil/	Galil DMC-4143 firmware and test-move logs (the controller is <i>not</i> installed — see §10)
../11rf/driveAmp/	Drive amplifier datasheet (KAW2051M12)
../11rf/arcDetector/	Arc detector product sheets and mechanical references
11rf/tests/11rf9Tests.tex	LLRF9 commissioning measurements (J. Sebek, 2021)

20.3 HVPS Source Material

Path	Content
hvps/documentation/wiringDiagrams/ei7307900000.pdf	EI-730-790-00-C0 / NWL #39308 — the transformer manufacturer's own schematic. Authoritative for the 2750 kVA / 127 A nameplate and all oil, pressure, vacuum and flow protection setpoints
hvps/documentation/schematics/sd7307900101.pdf	SD-730-790-01-C1 — HVPS system schematic
hvps/documentation/wiringDiagrams/wd7307940400.pdf	WD-730-794-04-C0 — contains the HVPS output voltage dividers
hvps/documentation/wiringDiagrams/wd7307900206.pdf	WD-730-790-02-C6 — master trigger enclosure wiring diagram
hvps/documentation/procedures/spear3HvpsHazards.tex	J. Sebek — stored energy, discharge time constants, klystron perveance
hvps/architecture/designNotes/EnerproVoltageandCurrentRegulatorBoardNotes.docx	J. Sebek — authoritative regulator board circuit analysis
hvps/architecture/designNotes/RFSystemMPSRequirements.docx	RF system MPS requirements
hvps/architecture/originalDocuments/slac-pub-7591.pdf	Cassel & Nguyen, PAC 1997 — crowbar timing, regulation, ripple
hvps/architecture/originalDocuments/ps3413600102.pdf	PS-341-360-01 — HVPS power section specification
hvps/controls/enerpro/enerproBoardHvps.docx	J. Sebek — installed Enerpro board revisions, serials, phase references
hvps/documentation/plc/CasselPLCCode.pdf	Full SLC-500 ladder logic listing, program SSRLV6-4-05-10
hvps/documentation/plc/plcNotesR1.docx	J. Sebek — PLC timing and the N7:10 setpoint filter
../hvps/documentation/procedures/	Maintenance and safety procedures, EWP's, lockout permits
../hvps/maintenance/	Stack installation checklists, phase tank maintenance

20.4 Switchgear and PPS Source Material

Path	Content
hvps/documentation/switchgear/gp3085000103.pdf	GP-308-500-01-R3 (LBL, 1977) — original design drawing; authoritative relay legend ; blocking relay blocks trip above 2000 A
hvps/documentation/switchgear/gp4397040201.pdf	GP-439-704-02-C1 — as-built schematic and sequence of operation
hvps/documentation/switchgear/id3088010601.pdf	ID-308-801-06-C1 — as-built connection wiring

Path	Content
hvps/documentation/switchgear/rossEngr713203.pdf	Ross 713203 E-1 — HCA-1-A driver and HQ3 contactor
pps/HoffmanBoxPPSWiring.docx	J. Sebek — terminal-by-terminal PPS wiring trace
pps/MSG\from\Jim\Sebek\to\Faya\about\PPS.md	2022 PPS concerns and upgrade drivers
pps/diagrams/README.md	PPS diagram note set and the PPS readback trace

20.5 Companion design documents

Document	Content
Designs/tex/docL-legacy-architecture/L_legacy_system_architecture.pdf	The legacy system reference. All legacy values in this report should agree with it
Designs/tex/docI-interlock-architecture/I_interlock_architecture.pdf	Legacy interlock and protection architecture
Designs/tex/docP-rf-physics/P_rf_physics_and_plant.pdf	RF physics, cavity parameters, beam loading
Designs/tex/docT-tuner-control/T_tuner_control_analysis.pdf	Tuner control system analysis